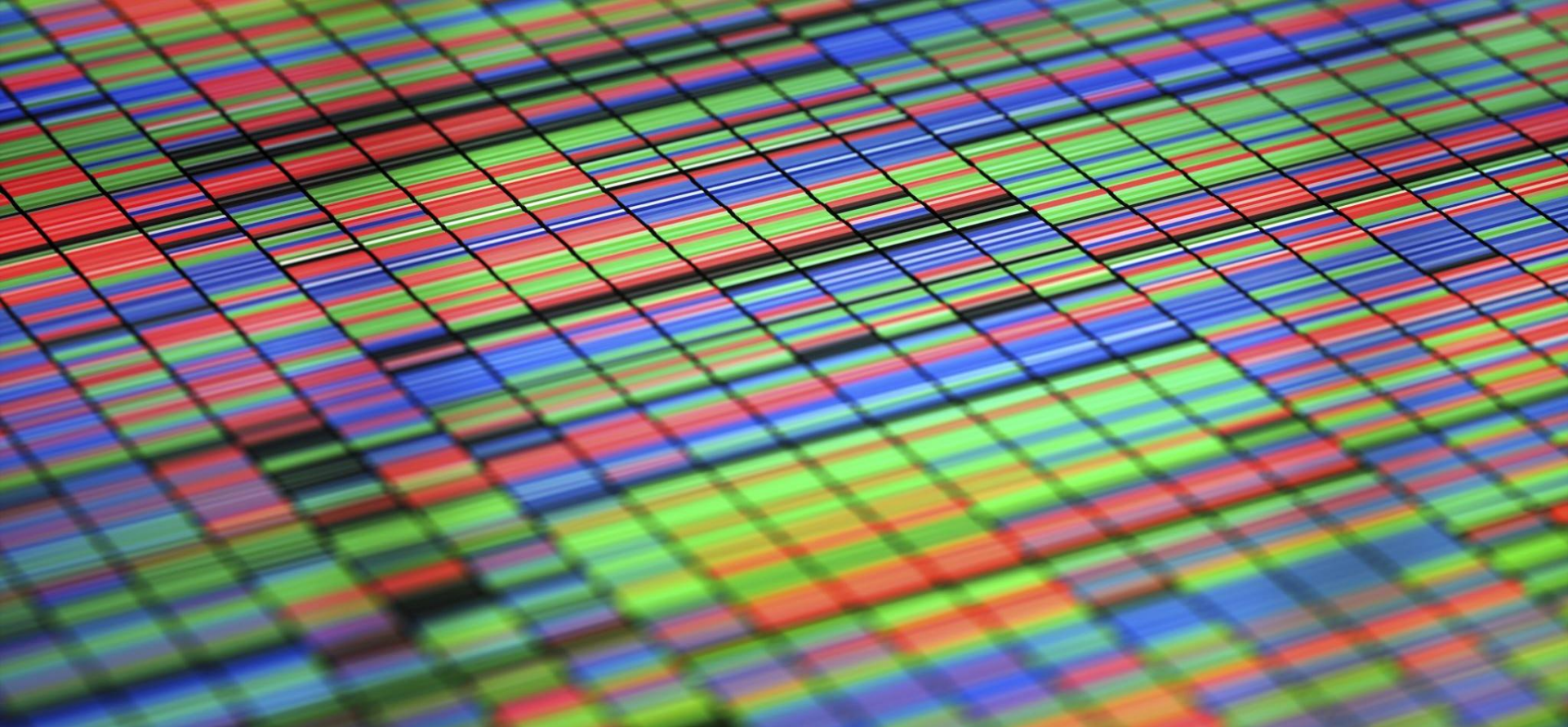




Data Warehouses

Lecture 14



Data Visualisation

Data Presentation

What is Business Data Visualization?

Data/information visualization

Create a mental imagery representation of data/information (meaning)

Business data

Abstract

Structured or semi-structured

Multidimensional

Complicated relationships

Directly comprehensible by human

Business data visualization

mainly for communication, analysis, and decision support

simple, abstract, direct

Why use visualization?

Visualization and BI

visualization is an important part of understanding

information seeking and decision making

Visualization tools have become increasingly important to business intelligence

people need support to make sense of and analyze complex data sets

Studies reveal:

75% of what we learn and remember comes from ocular stimulation

So...

the more visual a data presentation is

the more will be retained

Visualizations helps

Communicate (even hidden, additional) the meaning

Enhance problem solving capabilities

Ease the cognitive load of information processing

Recall or memorize data

Example

Credit card payment report

Legend

OK – on time

10 – 0 to 10 days late

20 – 10 to 20 days late

30 – 20 to 30 days late

Is it clear ?

How many missed the deadline by 10 to 20 days ?

Which year is the worst in terms of timely payments ?

Credit Card Payments Report

	1	2	3	4	5	6	7	8	9	10	11	12
2000	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2001	OK	OK	OK	OK	OK	30	30	30	30	10	10	OK
2002	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2003	OK	20	OK	10	OK	OK	OK	OK	OK	OK	OK	OK
2004	OK	OK	OK	OK	OK	OK	OK	30	OK	OK	OK	OK
2005	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2006	OK	OK	OK	OK	OK	10	OK	OK	OK	OK	OK	OK
2007	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2008	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2009	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK

	1	2	3	4	5	6	7	8	9	10	11	12
2000	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2001	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2002	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2003	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2004	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2005	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2006	OK	OK	OK	OK	OK	OK	10	OK	OK	OK	OK	OK
2007	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2008	OK	OK	OK	OK	OK	10	20	OK	10	20	OK	OK
2009	OK	10	OK	OK	30	30	OK	OK	OK	10	OK	OK

	1	2	3	4	5	6	7	8	9	10	11	12
2000	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2001	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2002	OK	OK	OK	OK	OK	OK	OK	OK	20	20	OK	OK
2003	OK	OK	OK	OK	10	OK	OK	30	20	OK	OK	OK
2004	OK	OK	OK	OK	OK	OK	OK	30	30	OK	OK	OK
2005	OK	OK	OK	OK	OK	OK	20	30	20	OK	OK	OK
2006	OK	OK	OK	OK	10	OK	OK	30	10	OK	OK	OK
2007	OK	OK	OK	OK	OK	30	20	30	10	OK	OK	OK
2008	OK	OK	OK	OK	OK	OK	30	10	OK	OK	OK	OK
2009	OK	OK	OK	OK	OK	OK	10	20	OK	OK	OK	OK

Legend:
 OK – "On Time"; 10 – "0 to 10 days late"; 20 – "10 to 20 days late"; 30 – "20 to 30 days late"

A better example

Conditional formatting

Legend

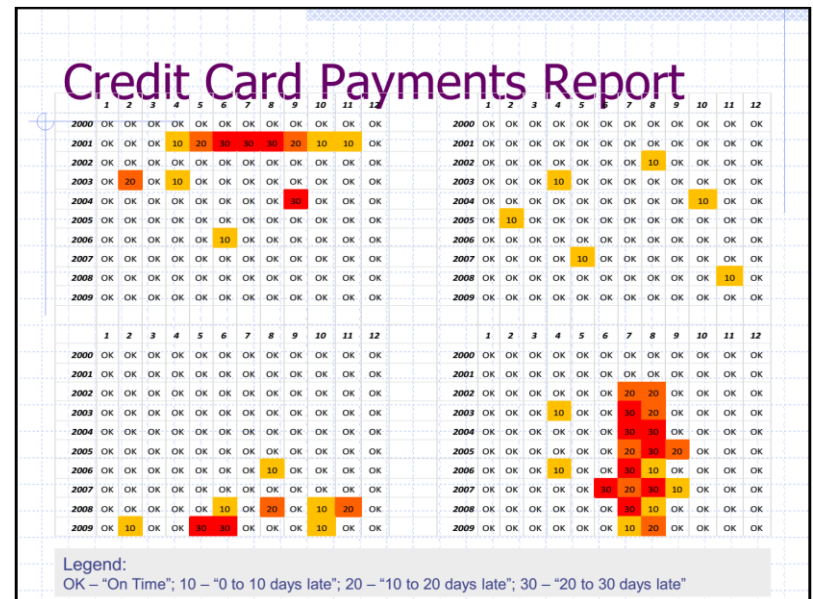
Same as before

The more red the cell is the more days late the client was

Is it clear ?

How many missed the deadline by 10 to 20 days ?

Which year is the worst in terms of timely payments ?



Use of visual graphics

Used correctly, visual communication tools and techniques can:

focus a group's attention, reinforce key points

help a discussion stay on the agenda

stimulate interest

improve message and comprehension

explain complex facts and processes

Example

Weather data
Different scope
50/18 years

Legend

Temperature
High/low
Record
Percipitation

Average Temperature													Years on Record: 50
°F	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	62	43	46	53	62	70	77	79	79	73	63	53	45
Average High Temperature													Years on Record: 50
°F	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	72	52	56	64	73	80	86	89	88	82	73	63	54
Average Low Temperature													Years on Record: 50
°F	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	52	33	36	43	51	59	67	70	69	64	53	42	36
Highest Recorded Temperature													Years on Record: 50
°F	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	105	79	80	89	93	95	101	105	102	98	95	84	79
Lowest Recorded Temperature													Years on Record: 50
°F	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	-8	-8	5	10	26	37	46	53	55	36	28	3	---
Average Precipitation													Years on Record: 50
in.	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	50.3	4.7	4.6	5.6	4.1	4	3.7	5.3	3.7	3.6	3	4	4.1

Average Temperature													Years on Record: 18
°F	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	62	43	46	53	63	71	75	78	78	71	64	54	45
Average High Temperature													Years on Record: 18
°F	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	67	48	51	59	70	77	81	84	84	76	68	59	50
Average Low Temperature													Years on Record: 18
°F	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	56	37	41	47	55	63	69	72	71	66	59	49	40
Highest Recorded Temperature													Years on Record: 18
°F	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	104	68	73	84	90	93	104	104	100	91	84	81	77
Lowest Recorded Temperature													Years on Record: 18
°F	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	21	25	27	30	37	49	55	57	55	50	36	32	21
Average Precipitation													Years on Record: 84
in.	YEAR	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
	38.8	0.3	0.5	0.8	1.9	3.4	4.3	9.5	10	4.9	1.8	0.6	0.2

Example

Weather data

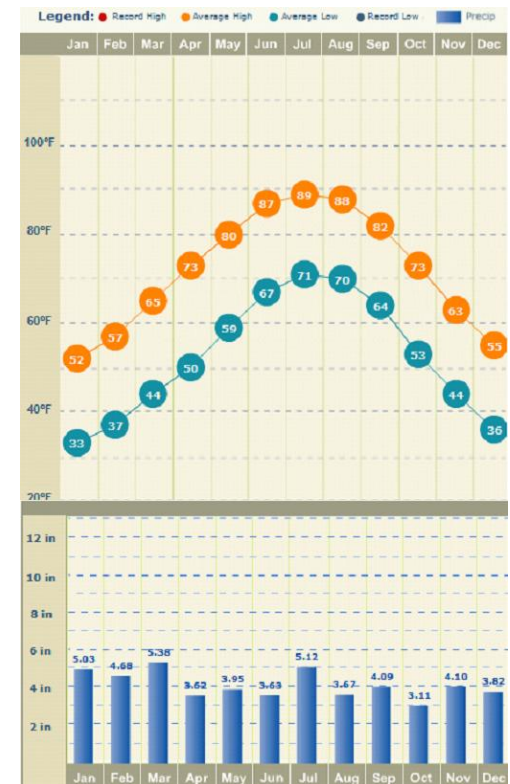
Different scope

60/18 years

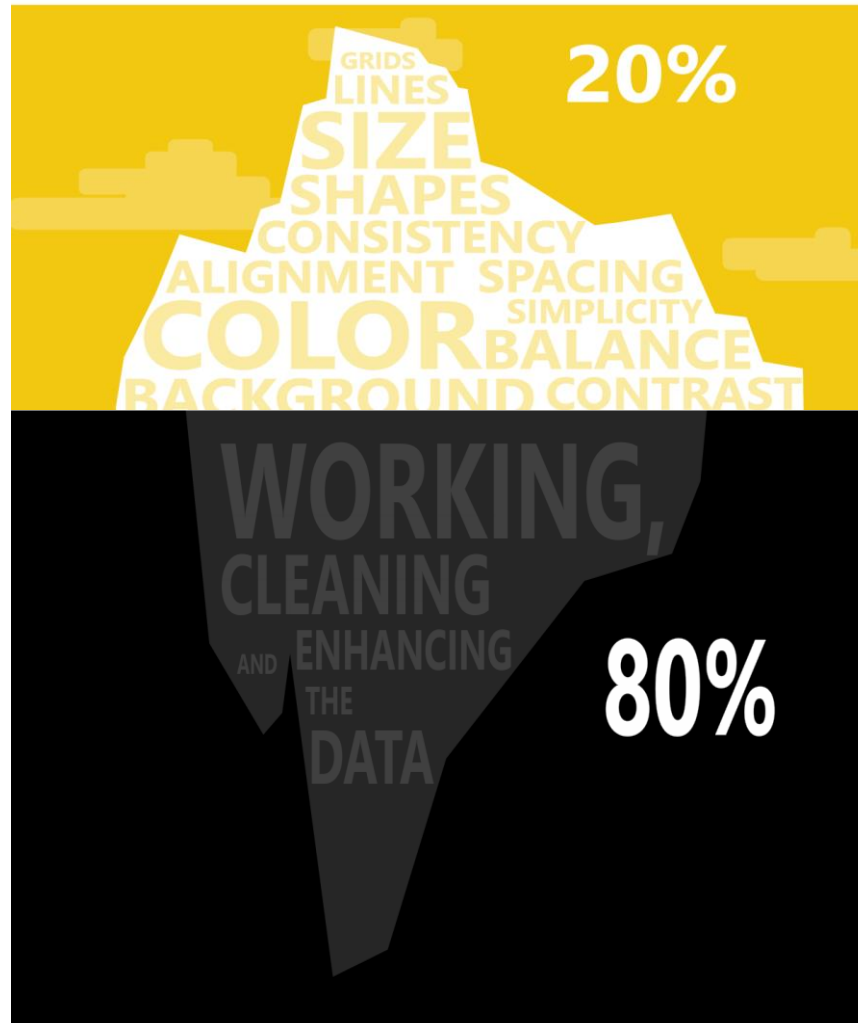
Legend

Temperature

Percipitation



Data visualisation Iceberg



4 Ways to Show Results



Reports



Gauges

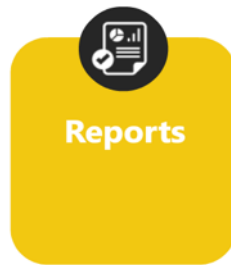


Dashboards



Infographics

Reports



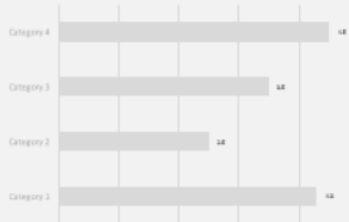
TEXT
TABLES
CHARTS
CARDS
INDICATORS
IMAGES
SLICERS

LOREM IPSUM DOLOR SIT AMET
CONSECTETUR ADIPISCING

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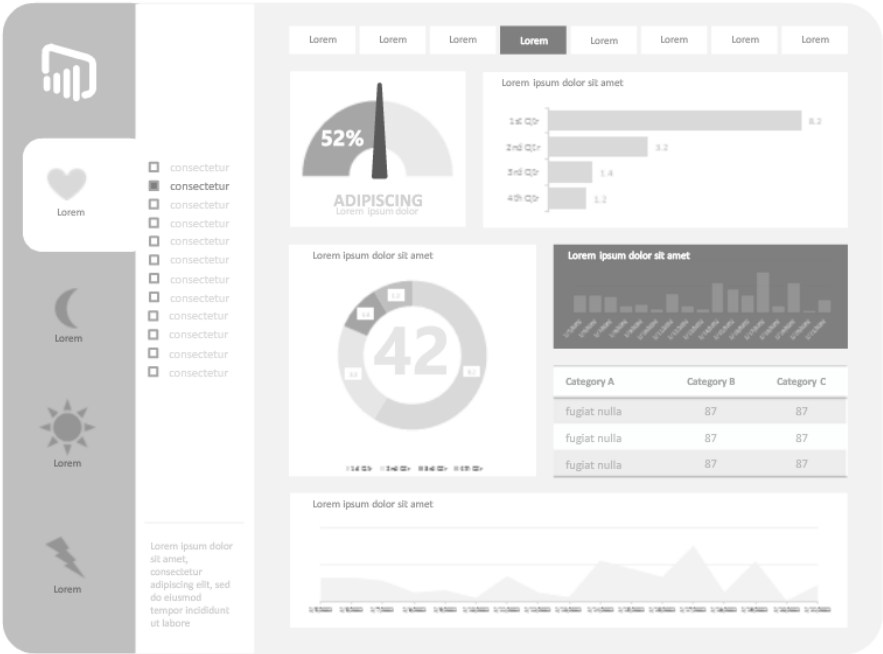
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Dashboards



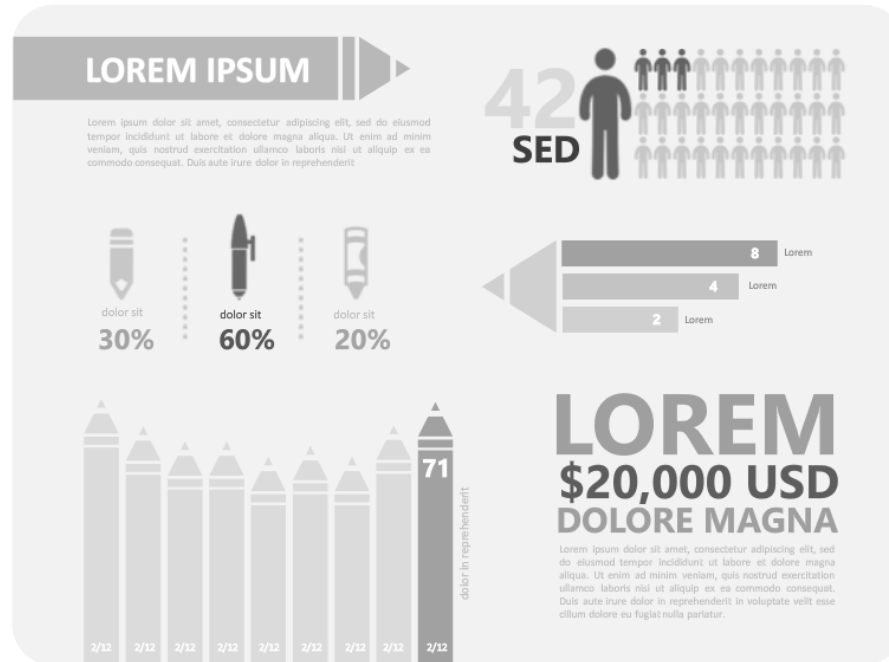
CHARTS
SLICERS
CARDS
INDICATORS
TABLES
IMAGES



Infographics



IMAGES
TEXT
CHARTS
CARDS
INDICATORS
SLICERS
TABLES



Gauges



INDICATORS
TABLES
CARDS
CHARTS
SLICERS
IMAGES



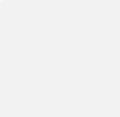
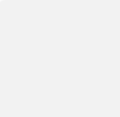
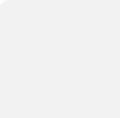
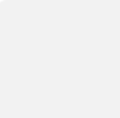
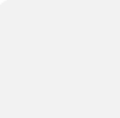
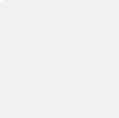
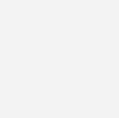
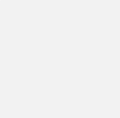
USERS

- CEO
- President
- Director

- Leader
- Manager
- Supervisor

- Analyst
- Marketing Specialist
- IT Technician

- Client
- Customer
- Employee



How to design a simple chart?

1. Message

Specify a message

2. Identifying the Statement Type

3. Selection of graphic elements

Selecting the type and specifying the type of graphic element

4. SSCOPe principle - Shape, Size, Color, Orientation, Position, tExture

Modification of a selected graphic element



First and foremost:

Understand the message you're trying to present with your data !

Your attention might focus on the overall sales trend from Jan. through May; how the dollar value of sales has changed over time.

In that case, your message would be that

„Sales have risen steadily since Jan.”

Sales by Product, \$

	A	B	C	Total
Jan.	88	26	7	121
Feb.	94	30	8	132
Mar.	103	36	8	147
Apr.	113	39	7	159
May	122	40	13	175



On the other hand, you might want to Focus on a single point in time.

Reading the figures across for May, for example, you might take note of the ranking of sales for Products A,B and C.

In that case, your message could be:

„In May, sales of Product A exceeded those of B and C by a wide margin.”

Sales by Product, \$

	A	B	C	Total
Jan.	88	26	7	121
Feb.	94	30	8	132
Mar.	103	36	8	147
Apr.	113	39	7	159
May	122	40	13	175

Looking at the same May data from yet another perspective, you might Focus on the percentage of total sales accounted for by each product.

Then your message might be:

„In May, Product A accounted for the largest share of total company sales.”

Sales by Product, \$

	A	B	C	Total
Jan.	88	26	7	121
Feb.	94	30	8	132
Mar.	103	36	8	147
Apr.	113	39	7	159
May	122	40	13	175
	70%	23%	7%	100%

First and foremost:

Understand the message you're trying to present with your data !

A few examples:

How not to do it ...

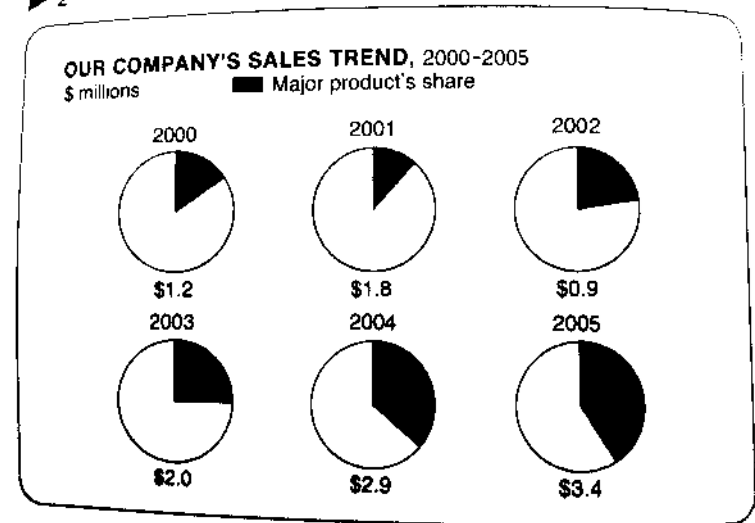
Idea:

“within the industry, our performance has been outstanding. For instance, our sales have grown, considerably since 2000, in spite of the decline in 2002 (result of a strike)”

Depicted:

A series of pie chart that show our major product's share increasing.

► 2



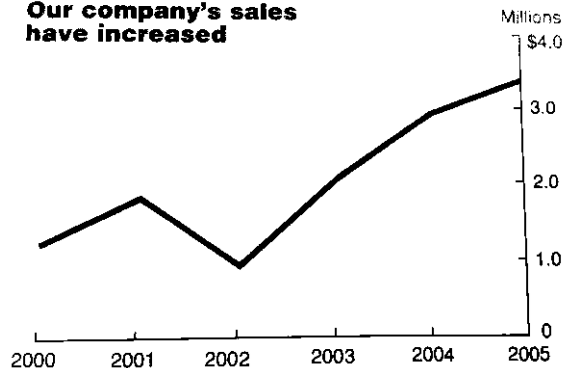
Idea:

“within the industry, our performance has been outstanding. For instance, our sales have grown, considerably since 2000, in spite of the decline in 2002 (result of a strike)”



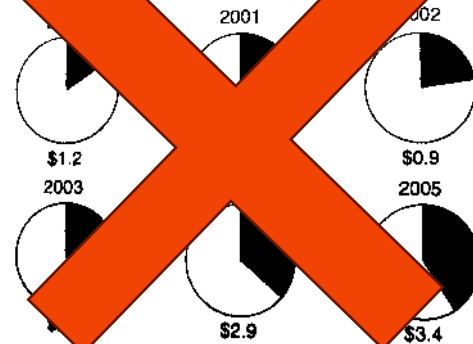
► 2

Our company's sales have increased



► 2

OUR COMPANY'S SALES TREND, 2000-2005
\$ millions



Idea:

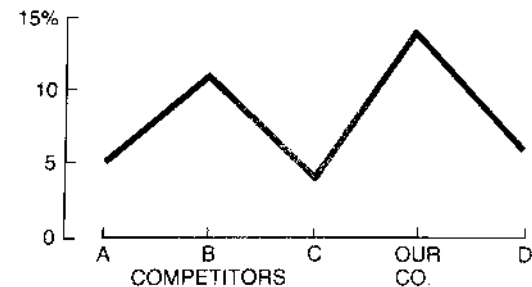
“Compared with our four major competitors, we rank first in return of investment with 14 percent ...”

Depicted

You read that the ROI is fluctuating

► 3

**COMPARISON OF 2005 RETURN
ON INVESTMENT, BY COMPANY**



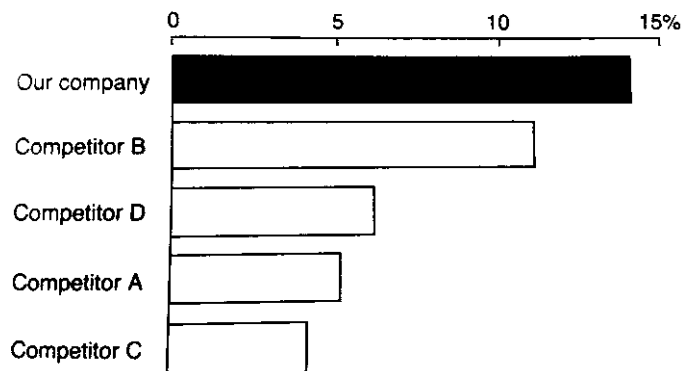
Idea:

“Compared with our four major competitors, we rank first in return of investment with 14 percent ...”



► 3

In 2005, our company ranks first in return on investment



► 3

COMPARISON OF 2005 RETURN ON INVESTMENT, BY COMPANY



Idea:

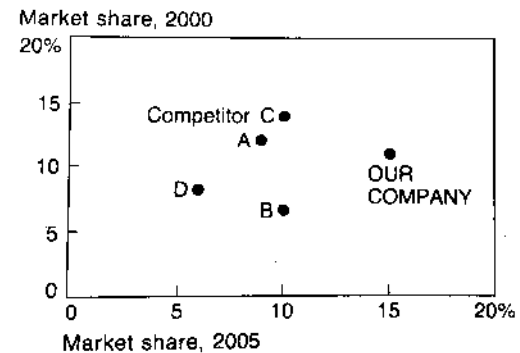
“... and our market share has increased since 2000 along with that of one competitor, while the other three lost share.”

Depicted:

Relation between market share in 2000 and 2005

► 4

MARKET SHARE TRENDS, BY COMPANY
2000-2005



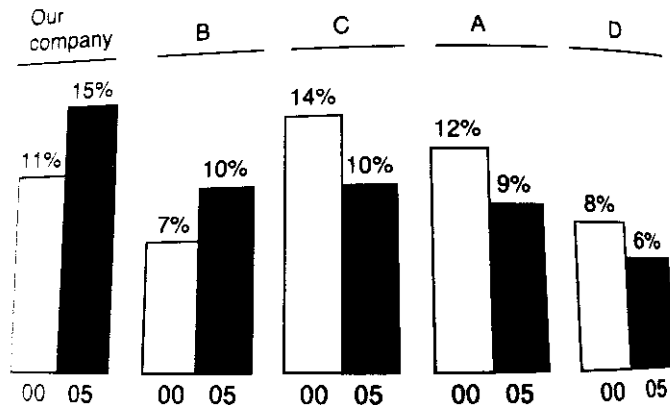
Idea:

“... and our market share has increased since 2000 along with that of one competitor, while the other three lost share.”



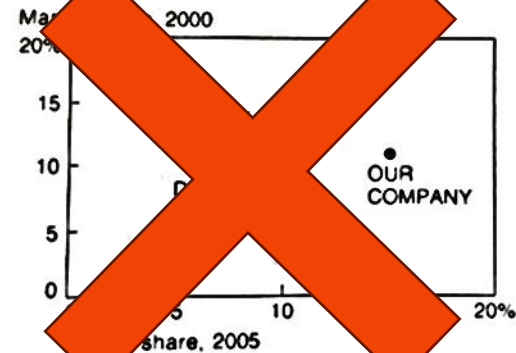
► 4

Our company's market share has improved since 2000



► 4

**MARKET SHARE TRENDS, BY COMPANY
2000-2005**



Process

First

Understand the message you're trying to present with your data

Second,

Identify the comparison

Five basic kinds of comparison:

Component – percentage of total

Item – ranking of items

Time series – changes over time

Frequency distribution – items within ranges

Correlation – relationship between variables

Third,

chart

Component comparison

Interested primarily in showing the size of each part as a percentage of the total

Example:

In may, Product A accounted for the largest share of total in company sales

Client share of market in 2005 is less than 10 percent of the industry

Any time the message contains words such, as

Share, percentage of total, accounted for X percent,

Item comparison

Compare how things rank

Are they about the same, or is one more or less than the other

Example:

In May, sales of Product A exceeded those of Products B and C

Client's return on sales ranks fourth

Words indicating

Larger than, smaller than, equal

Time series comparison

Not interested in the size of each part in a total or how they are ranked

Interested in how they change over time

Example,

Sales have risen steadily since January

ROI has decreased sharply over the past five years

Words

Change, grow, risen, decline, increase, decrease, fluctuate

Frequency distribution comparison

Shows how many items fall into a series of progressive numerical ranges

Example

In May, most sales were in the 1000 to 2000 range

The majority of shipments are delivered in five to six days

Words

X to y range, concentration, frequency, distribution

Application

Generalizing likely events on the basis of a sample of observations

predict risk, probability, or chance

25% chance that shipment will take 5 or less days

Summarizing vast amounts of data to demonstrate some meaningful relationship

Number of employees against different salary ranges

Correlation comparison

Shows weather the relationship between two variables follows (or fails) the pattern you would expect

Example:

Sales performance in May shows no relationship between sales and the salesperson's experience

CEO compensation does not vary with size of company

Words

Related to, increased with, decreased with, changes with, varies with, (or don't's)

Match

- A. Component Comparison
- B. Item Comparison
- C. Time Series Comparison
- D. Frequency Distribution
- E. Correlation Comparison

Sales are forecast to increase over the next 10 years

The largest number of employees ears between \$30000 and \$35000

Higher price of gasoline brands does not indicate better performance

In September, the turnover rates for the six divisions were about the same

The sales manager spends only 15% of his time in the field

Size of merit increases is not related to tenure

Last year, most turnover was in the 30 to 35 age group

Region C ranks last in productivity

Our company's earnings per share is declining

The largest share of total funds is allocated for manufacturing

There is a relationship between profitability and compensation

In August, two plants out produced the other six by a wide margin

Match – results

A. Component Comparison

5, 10

B. Item Comparison

4, 8, 12

C. Time Series Comparison

1, 9

D. Frequency Distribution

2, 7

E. Correlation Comparison

3, 6, 11

1. Sales are forecast to increase over the next 10 years
2. The largest number of employees ears between \$30000 and \$35000
3. Higher price of gasoline brands does not indicate better performance
4. In September, the turnover rates for the six divisions were about the same
5. The sales manager spends only 15% of his time in the field
6. Size of merit increases is not related to tenure
7. Last year, most turnover was in the 30 to 35 age group
8. Region C ranks last in productivity
9. Our company's earnings per share is declining
10. The largest share of total funds is allocated for manufacturing
11. There is a relationship between profitability and compensation
12. In August, two plants out produced the other six by a wide margin

Determine the message

From data to message

Key to choosing the appropriate chart form

be clear about the specific point you want to make, what you want to show

Identify the comparison

From message to comparison

The message will imply one of the five basic kinds of comparison:

Component, item, time series, frequency distribution or correlation

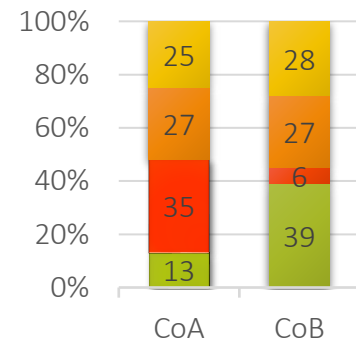
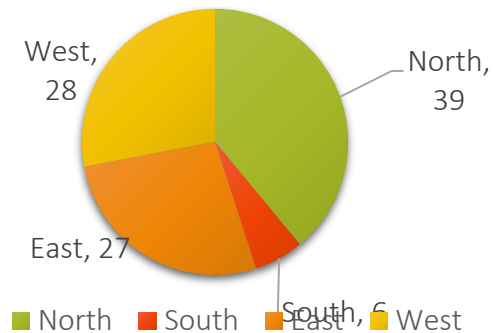
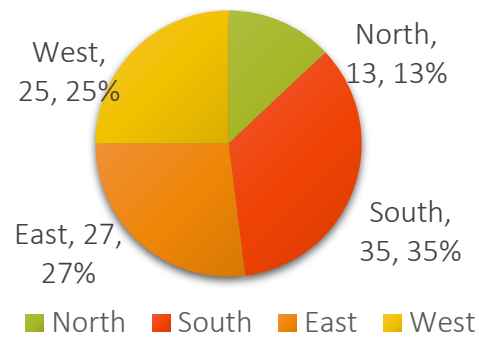
Select the chart form

From comparison to chart

Each comparison will lead to one of the five chart forms

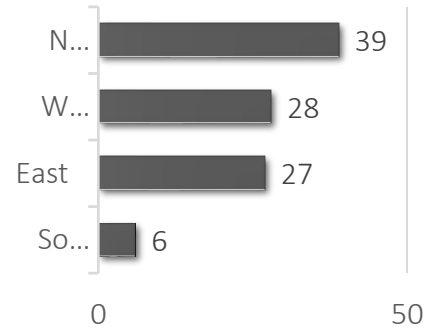
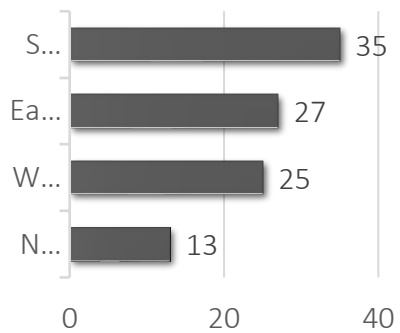
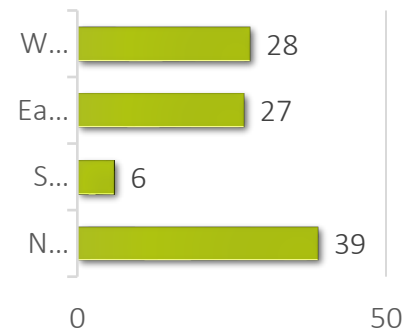
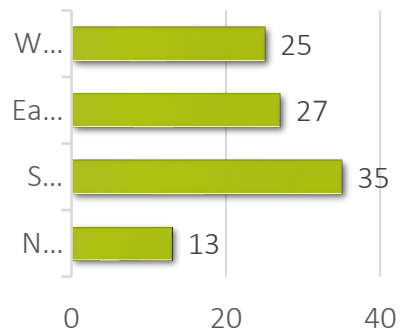
What is depicted?

	Co A	Co B
North	13	39
South	35	6
East	27	27
West	25	28



What is depicted?

	Co A	Co B
North	13	39
South	35	6
East	27	27
West	25	28



What is depicted?

	Co A	Co B
North	13	39
South	35	6
East	27	27
West	25	28

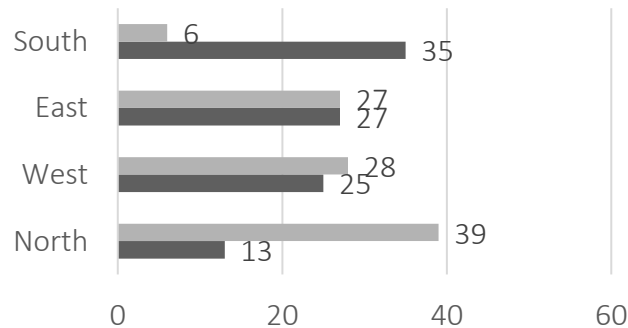
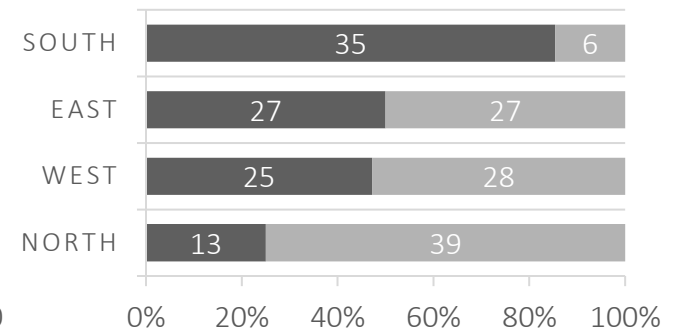
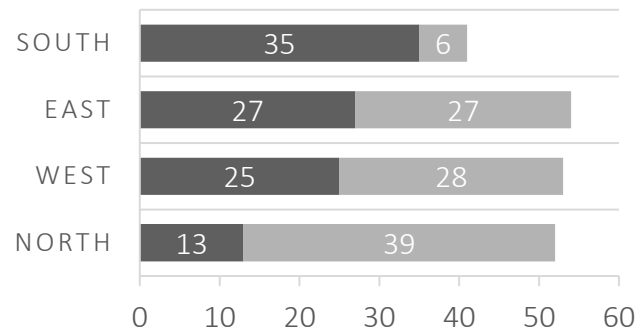


Chart type matters

Once you understand what message you're trying to send with the data

it's time to select the best method for displaying that information.

Different chart types cater best to different methods

Essentially when it comes to quantitative charts, there are only 5 basic chart forms:

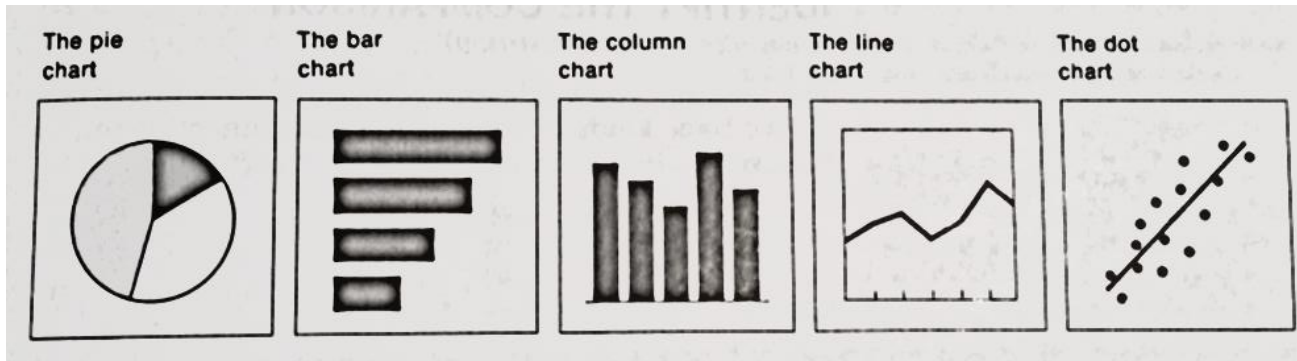
Pie chart

Bar chart

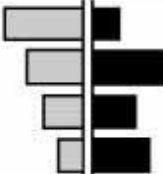
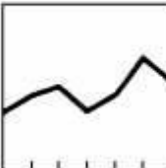
Column chart

Line chart

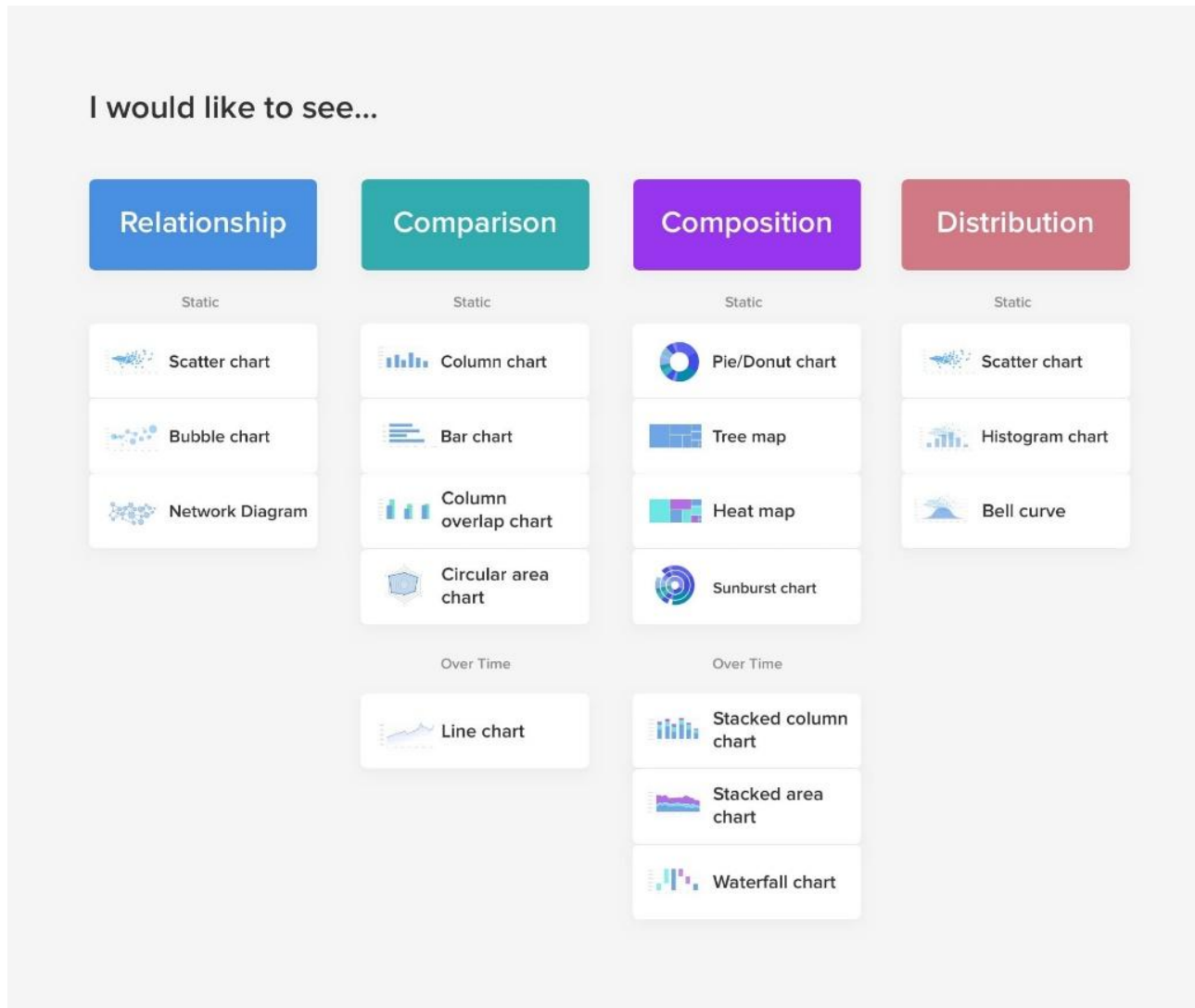
Dot chart



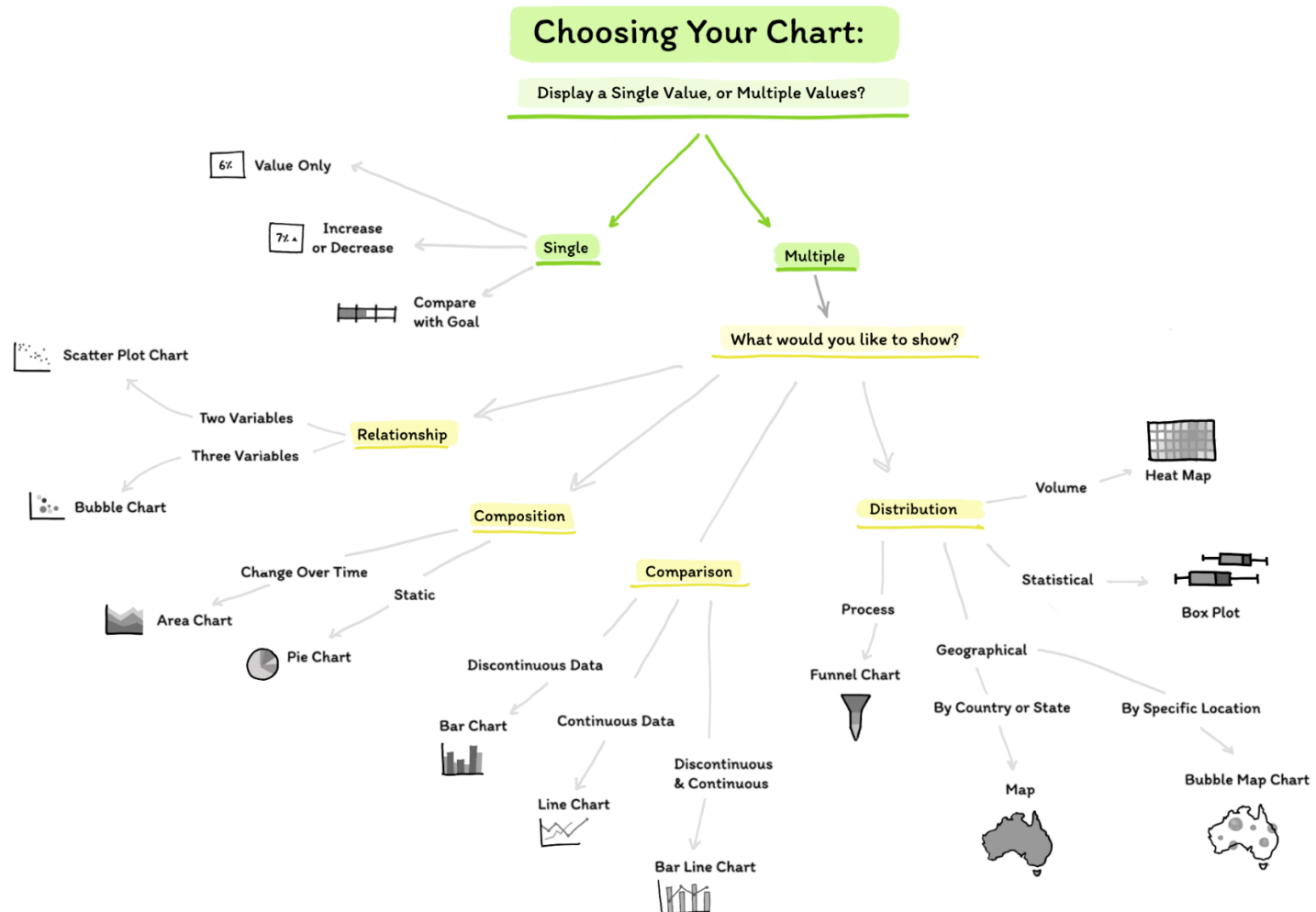
BASIC CHART FORMS

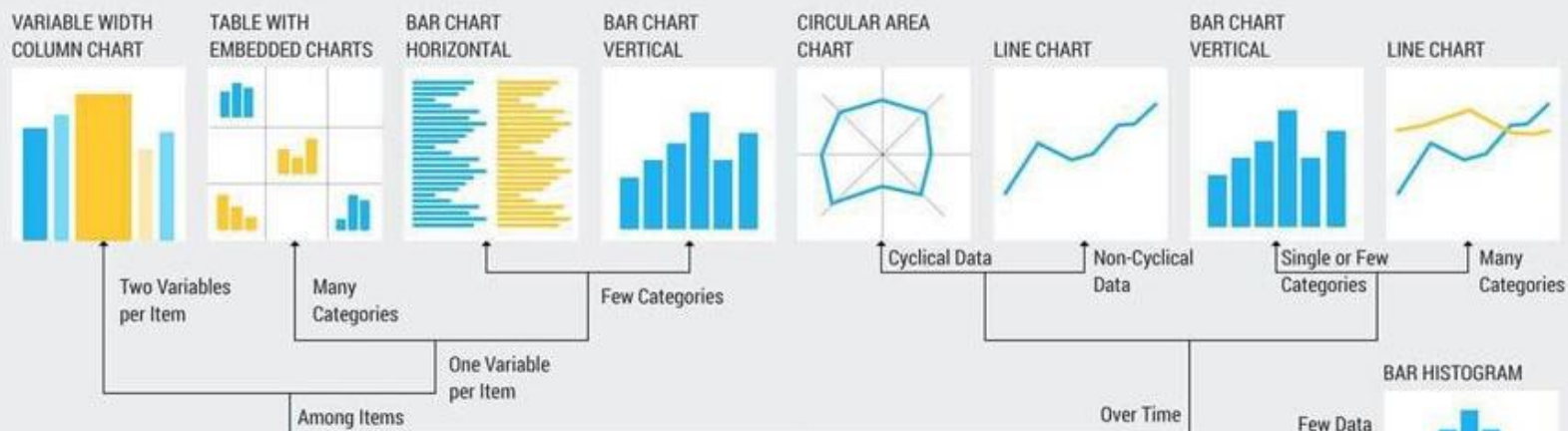
KINDS OF COMPARISON					
	COMPONENT	ITEM	TIME SERIES	FREQUENCY	CORRELATION
PIE					
BAR					
COLUMN					
LINE					
DOT					

There are other proposals:



There are other proposals:





SCATTER PLOT



SCATTER PLOT BUBBLE SAZE



Two Variables

Three or more Variables

RELATIONSHIP

COMPARISON

What would you like to show?

COMPOSITION

DISTRIBUTION

Few Data Points

Single Variable

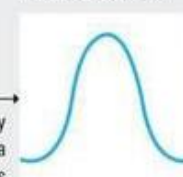
Many Data Points

Two Variables

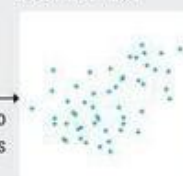
BAR HISTOGRAM



LINE HISTOGRAM



SCATTER PLOT



Changing Over Time

Static

Few Periods

Many Periods

Only Relative Differences Matter

Relative and Absolute Differences Matter

Only Relative Differences Matter

Relative and Absolute Differences Matter

Simple Share of Total

Accumulation or Subtraction to Total

Components of Components

Accumulation to total and absolute difference matters



STACKED 100% BAR CHART



STACKED BAR CHART



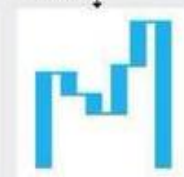
STACKED AREA 100% CHART



STACKED AREA CHART



PIE CHART



WATERFALL CHART



STACKED 100% BAR CHART WITH SUB-CATEGORIES



TREE MAP

Visual Vocabulary

Inspired by Jon Schwabish and Severino Ribeca continuum

<https://ft-interactive.github.io/visual-vocabulary/>

Visual Vocabulary

Designing with data

There are so many ways to visualise data – how do we know which one to pick? Click on the coloured categories below to decide which data relationship is most important in your story, then look at the different types of chart within the category to form some initial ideas about what might work best. This list is not meant to be exhaustive, nor a wizard, but is a useful starting point for making informative and meaningful data visualisations

Inspired by the Graphic Continuum by Jon Schwabish and Severino Ribeca

Deviation Correlation Change v Time Ranking Distribution Part to whole Magnitude Spatial Flow

Flow

Show the reader volumes or intensity of movement between two or more states or conditions. These might be logical sequences or geographical locations

Examples of use

Movement of funds, trade, migrants, lawsuits, information, relationship graphs

Chart types

sankey



Shows changes in flows from one condition to at least one other, good for tracing the eventual outcome of a complex process

waterfall



Designed to show the sequencing of data through a flow process, typically budgets. Can include v.t. components

chord



A complex but powerful diagram which can illustrate 2-way flows (and net winner) as a matrix

network



Used for showing the complexity and inter-connectiveness of relationships of varying types

Used to show the location of individual events/features - make sure to annotate any patterns the reader should see. (declined)

Grid based data values mapped with an intensity colour scale. As choropleth map - but not mapped to an administrative unit.

that makes sense to reader

highlighting values

Simplified methodology

There is also an interesting simplified methodology – based on the finding that 90% of cases, dashboards contain data of three types:

Ranking

Quantitative comparison based on the “more-less” principle: who is in first place, who is in the top 5 and by what margin, and who is lagging behind by how much.

For example, employee rankings by sales volume, article expenditure rankings, branch rankings by headcount.

The basic Fig. for visualizing such data is a column (horizontal or vertical).

Dynamics

This is the change in indicators over time (year, month, day) or other ordered periods, such as project lifecycle stages.

It is important to show the overall trend: growth, decline, or seasonal fluctuations in the indicator.

The basic Fig. for dynamics is a line on a timeline axis (usually called a timeline). Also, a column chart is suitable for this purpose.

Structure

Analysis of the composition of the whole, where the emphasis is not on quantity, but on percentages.

It helps to understand which segments and categories contribute the most to the overall result. For example, sales structure by markets, expenditure structure by projects.

The basic Fig. is a circle divided into sectors. Hence the metaphor of “pie” and “donut”.

Visualisation Compass

After you have determined the data type,
you need to understand how many categories you want to show:
5, 10, 15, or 50.

Metaphorical compass.

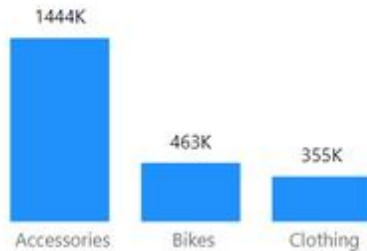
Depending on how many categories you have, the arrow will indicate which chart is best suited for you.

NOTE – If you want to show 50 and more categories
then no chart will help you.

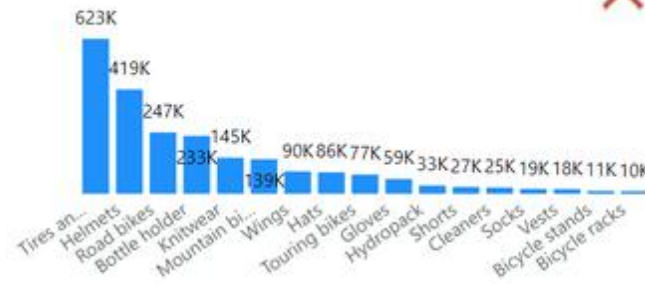


Ranking

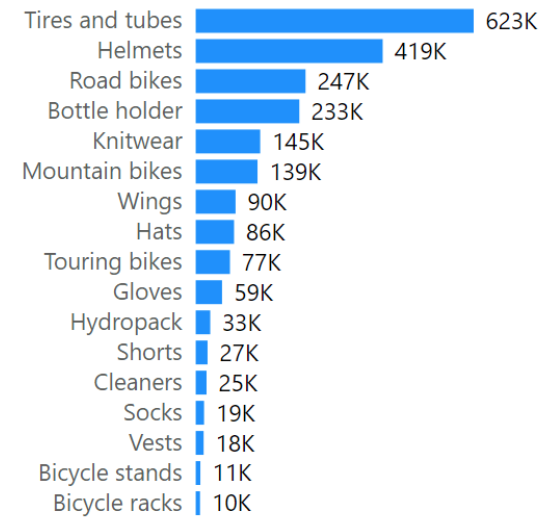
Sales by Product category, \$



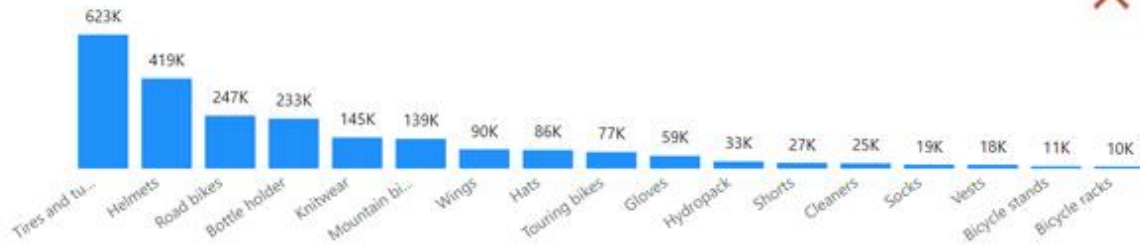
Sales by Product subcategory, \$



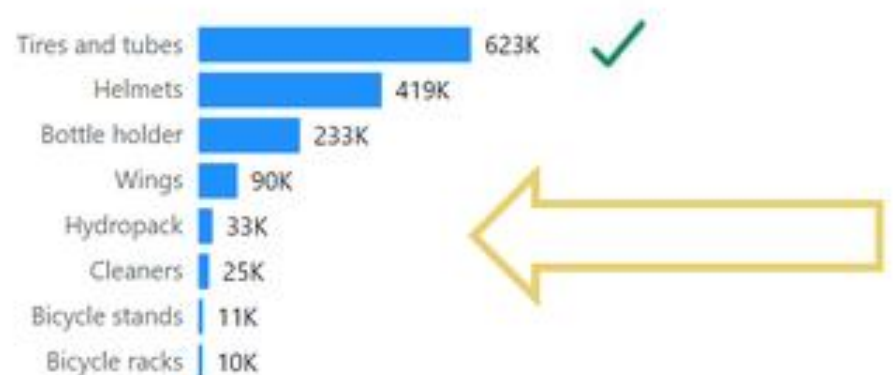
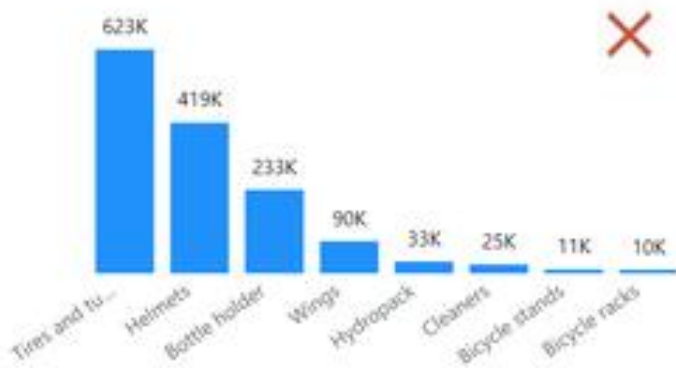
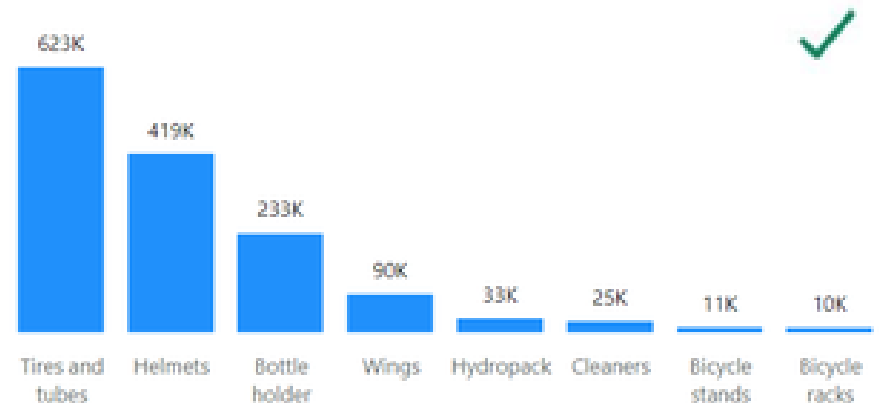
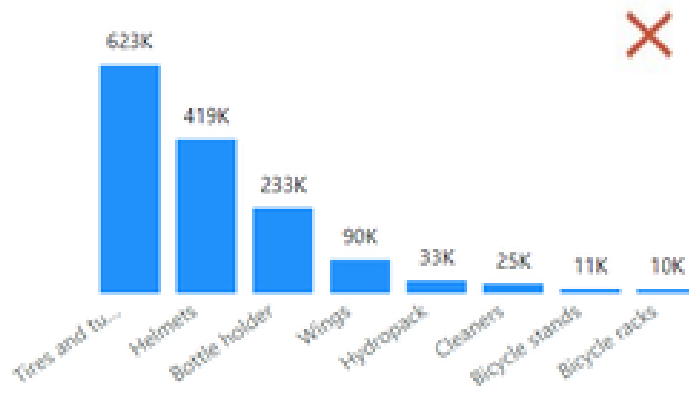
Sales by Product subcategory, \$



Sales by Product subcategory, \$

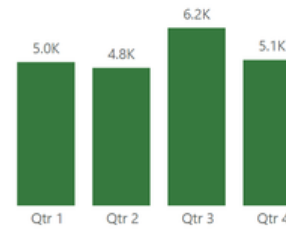


Ranking



Dynamics

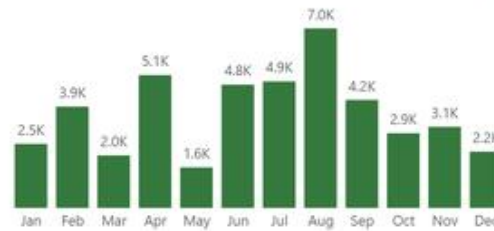
Sales by Quarter, \$



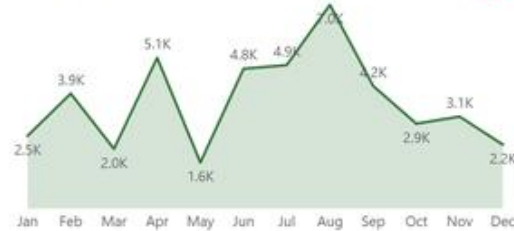
Sales by Day, \$



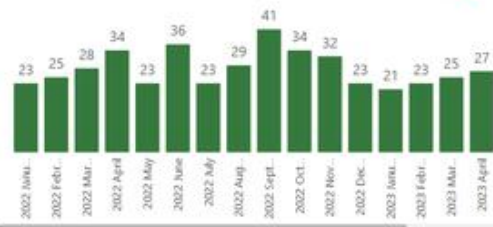
Sales dynamics, \$



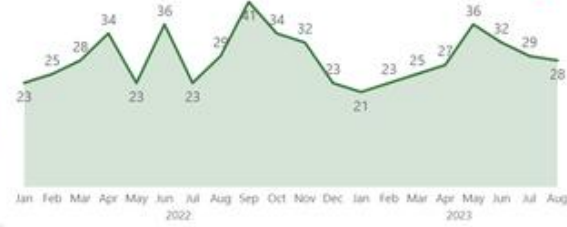
Sales dynamics, \$



Sales dynamics, \$



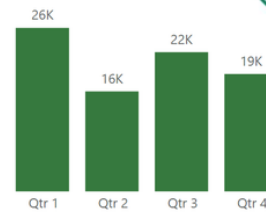
Sales dynamics, \$



Sales dynamics, \$

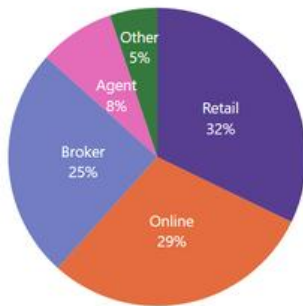


Sales dynamics, \$

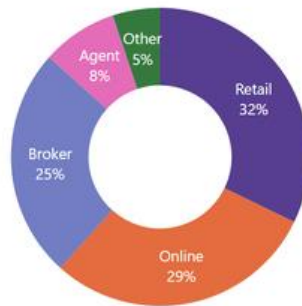


Structure

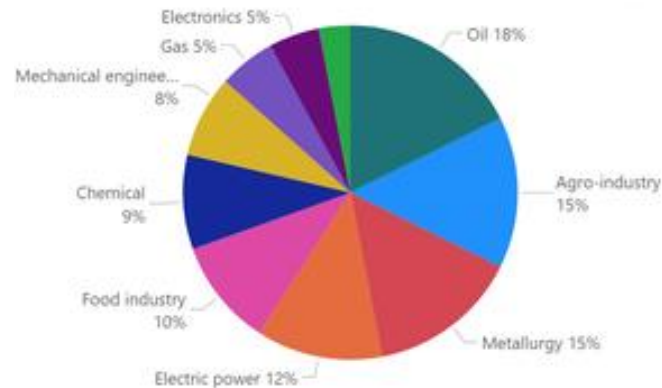
Distribution Channels



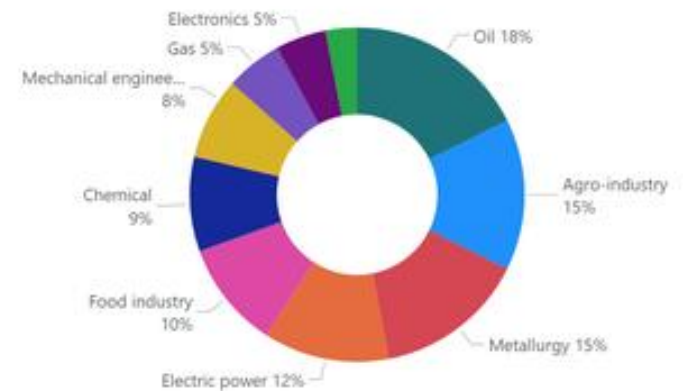
Distribution Channels



Export Structure

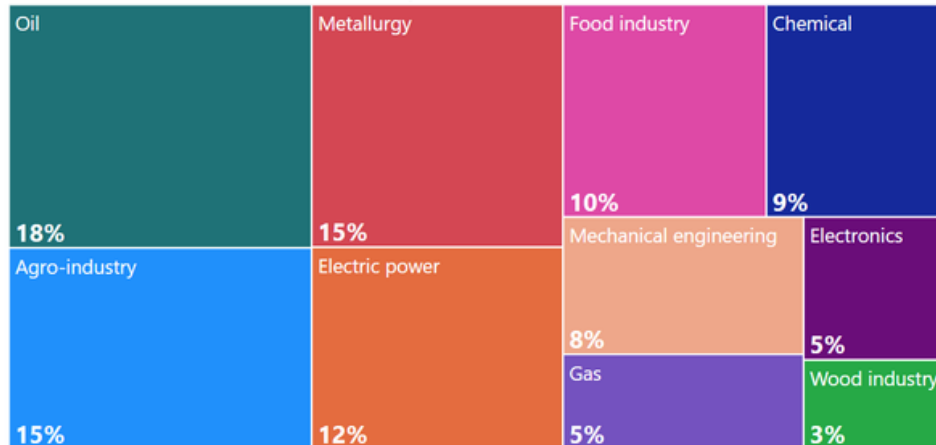


Export Structure

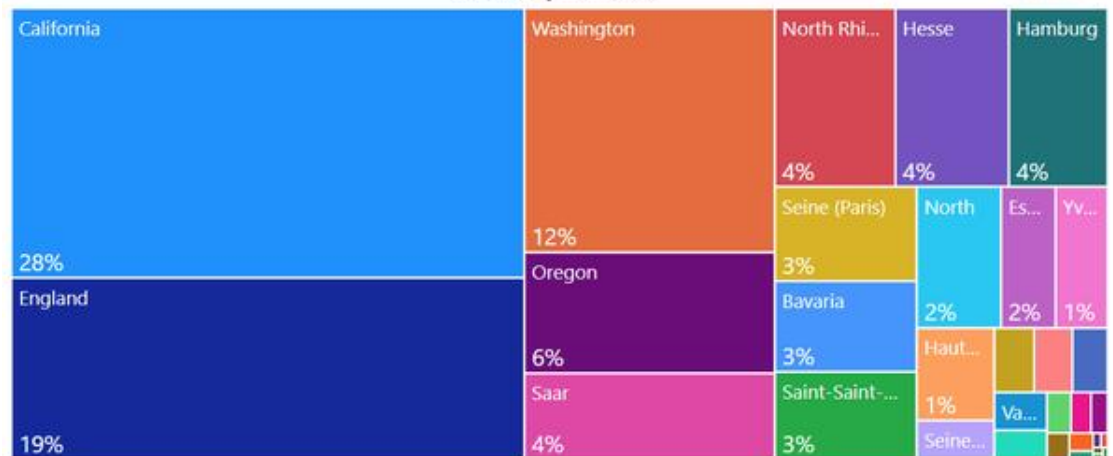


Structure

Export Structure

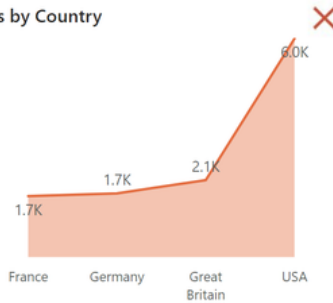


Sales by State, \$

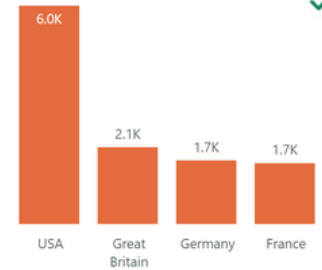


Typical mistakes

Sales by Country



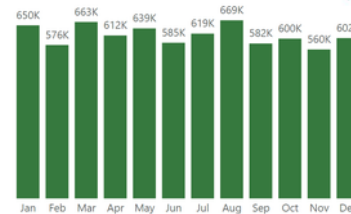
Sales by Country



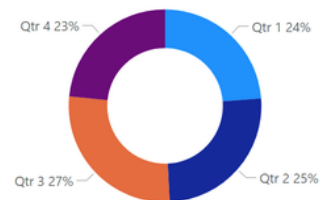
Sales dynamics, \$



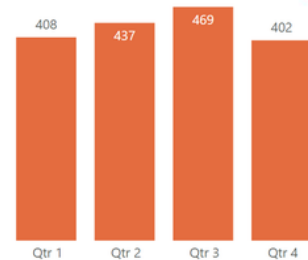
Sales dynamics, \$



Sales by Quarter



Sales by Quarter



How to design a visualization?

Basic consideration

Choice of visual properties

SSCOPE

Shape, Size, Color, Orientation, Position, tExture

Choice of visual form (visual elements)

Embedded visuals

Visual cues, conditional formatting, Sparkline

Standalone visuals

Charts, Illustrational diagrams, Infographics

Embedded visuals

Conditional formatting

Direct formatting on text or numbers using visual properties

Sparkline

a small chart embedded in a context of words, numbers, tables, images, or other type of information.

presents the general shape of the variation in a simple and highly condensed way.

Charts

unique combination of symbols with visual properties which directly represents quantitative values

Visual Clues

Additional graphical elements

E.g. icons

		2010	2011	Percentage change 2011 on 2010	
ECONOMIC PERFORMANCE					
Automotive manufacturing sector turnover	(£ billion)	49	55	12.2	●
Expenditure on business R&D	(£ billion)	1.3	1.3	0.0	●
Total number of cars and CVs produced	(million) (UK) (WI)	1.39	1.46	5.1	●
Total new cars and CV registrations	(million) (UK) (WI)	2.30	2.25	-2.3	●
Signatories' combined turnover	(£ billion) (AS)	42.3	45.3	7.0	●
Total number of vehicles produced	(million) (AS)	1.30	1.38	6.0	●
ENVIRONMENTAL PERFORMANCE					
Production inputs					
Total combined energy use	(GWh) (AS)	4,659	4,381	-6.0	●
Energy used per vehicle produced	(MWh/unit) (VMs)	2.6	2.2	-14.0	●
Total combined water use	(000m ³) (AS)	5,468	4,875	-10.8	●
Water use per vehicle produced	(m ³ /unit) (VMs)	3.2	2.8	-14.9	●
Material output					
Total combined CO ₂ equivalents	(tonnes) (AS)	1,442,896	1,395,233	-3.3	●
CO ₂ equivalents per vehicle produced	(tonnes/unit) (VMs)	0.8	0.7	-13.3	●
Volatile Organic Compounds emissions (cars)	(g/m ²) (VMs)	36	35	-0.6	●
Volatile Organic Compounds emissions (vans)	(g/m ²) (VMs)	64	61	-4.1	●
Total combined waste to landfill	(tonnes) (AS)	18,302	13,971	-23.7	●
Waste to landfill per vehicle produced	(kg/unit) (VMs)	8.8	7.1	-19.8	●
Vehicle use					
Average new car CO ₂ emissions	(g/lkm) (AC)	144.2	138.1	-4.2	●
SOCIAL PERFORMANCE					
Number of jobs dependent on the sector	('000) (WI)	737*	719*	-2.4	●
Combined number of employees	(AS)	67,791	74,155	9.4	●
Number of lost-time incidents	(AS)	198	180	-9.1	●
Number of training days per employee	(AS)	2.93	2.94	0.3	●
Number of new apprentices (retail, engineering)	(WI)	27,450**	30,650**	11.7	●

(WI) Whole industry data, (AC) All car sales in the UK, (AS) All signatories, (VMs) UK vehicle manufacturer signatories, (CV) Commercial vehicles, (CO₂) Carbon Dioxide.

The report has 17 signatories which represent 94.1% of vehicle production in the UK.

Conditional Formatting

Changes in formatting of certain values

Depending on the context

Depending on the value

Device	Release Date	Data Connections/sub	UL Data Volume (kbits/sub)	DL Data Volume (kbits/sub)	Operating System
iPhone 3G	Jun/08	100%	100%	100%	iOS
iPhone 4	Jun/10	122%	232%	223%	iOS
iPhone 4S	Oct/11	167%	306%	279%	iOS
iPhone 5	Sep/12	165%	342%	414%	iOS
HTC Desire	Mar/10	157%	158%	186%	Android
HTC Desire S	Apr/11	190%	283%	164%	Android
HTC Sensation XL	Nov/11	240%	264%	317%	Android
HTC One X	May/12	195%	284%	286%	Android
Samsung Galaxy S	Jun/10	95%	94%	118%	Android
Samsung Galaxy S II	Feb/11	125%	183%	172%	Android
Samsung Galaxy S III	Sep/12	110%	385%	315%	Android
Samsung Galaxy Note 2	Sep/12	114%	373%	307%	Android
Sony Ericsson Xperia X10i Mini Pro	Mar/10	157%	96%	114%	Android
Sony Ericsson Xperia Arc	Apr/11	138%	200%	215%	Android
Sony Ericsson Xperia S	Feb/12	122%	212%	252%	Android
Nokia N800	Jan/07	62%	51%	53%	Maemo
Nokia E72	Jun/09	71%	37%	38%	Symbian
Nokia E7	Feb/11	74%	59%	61%	Symbian
Blackberry Bold	Nov/10	106%	68%	70%	Blackberry OS
Blackberry Bold Touch	Sep/11	84%	150%	136%	Blackberry OS
Blackberry Curve	May/12	100%	113%	137%	Blackberry OS
3G Modems	(various)	14%	2439%	2222%	(various)
iPad	Apr/10	128%	229%	261%	iOS
iPad 2	Mar/11	125%	287%	259%	iOS
Samsung Galaxy Tab 2 10.1	May/12	48%	232%	309%	Android

Sparkline

Small embedded graphs

In a certain context

To aid quick depiction of a general shape

Simple or complex

Team	W/L By Week (H=Dark)	TO Ratio	Pts/G vs Avg	Yds/G
Tennessee Titans		14	23.4	313.6
New York Giants		9	26.7	355.9
Pittsburgh Steelers		4	21.7	311.9
Indianapolis Colts		9	23.6	335.5
Carolina Panthers		6	25.9	349.7
Atlanta Falcons		-3	24.4	361.2
Baltimore Ravens		13	24.1	324.0
New England Patriots		1	25.6	365.4
Miami Dolphins		17	21.6	345.6
Minnesota Vikings		-6	23.7	330.9
Philadelphia Eagles		3	26.0	350.5
Tampa Bay Buccaneers		4	22.6	341.0
Arizona Cardinals		0	26.7	365.8
New York Jets		-1	25.3	331.7
Dallas Cowboys		-11	22.6	344.5
Chicago Bears		5	23.4	295.9
New Orleans Saints		-4	28.9	410.7
San Diego Chargers		4	27.4	349.0
Washington Redskins		0	16.6	320.0
Denver Broncos		-17	23.1	395.8
Houston Texans		-10	22.9	382.1
Buffalo Bills		-8	21.0	305.1
San Francisco 49ers		-17	21.2	311.1
Green Bay Packers		7	26.2	351.1
Oakland Raiders		1	16.4	272.2
Jacksonville Jaguars		-7	18.9	319.1
Cleveland Browns		5	14.5	249.1
Cincinnati Bengals		-2	12.8	245.4
Seattle Seahawks		-7	18.4	274.1
Kansas City Chiefs		5	18.2	308.7
St. Louis Rams		-5	14.5	287.2
Detroit Lions		-9	16.8	268.2

Charts

Unique combination of symbols

Certain convention

Designed to represent numerical values

Quantitative analysis

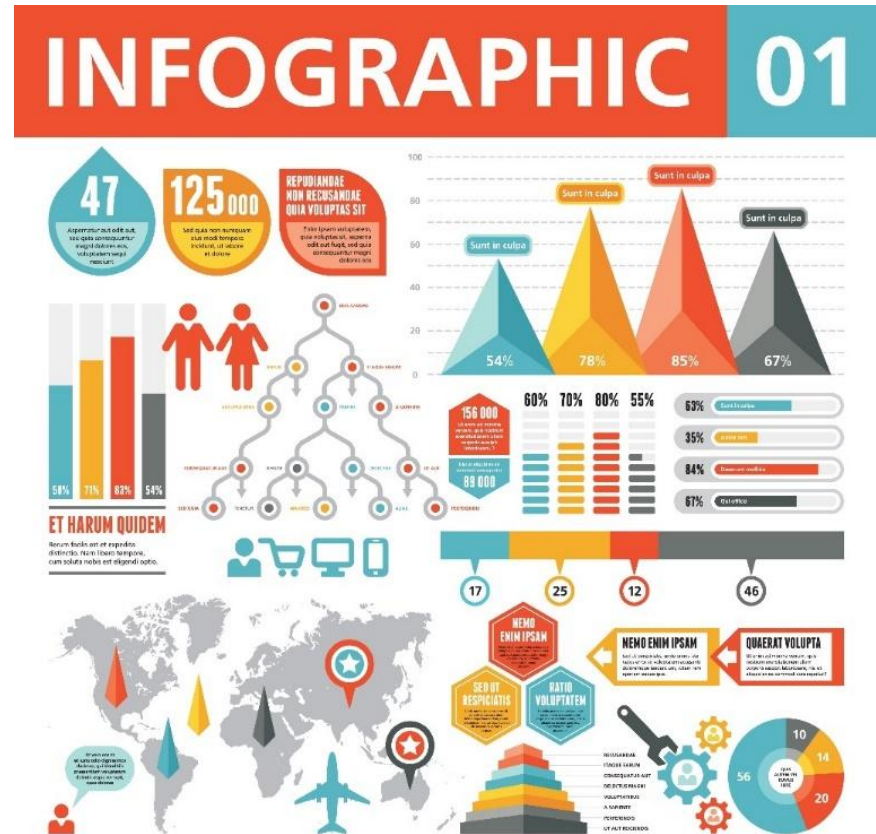


Infographics

Graphical representation of selected information, data and/or knowledge

Oriented towards fast and clear depiction

Designed to aid perception

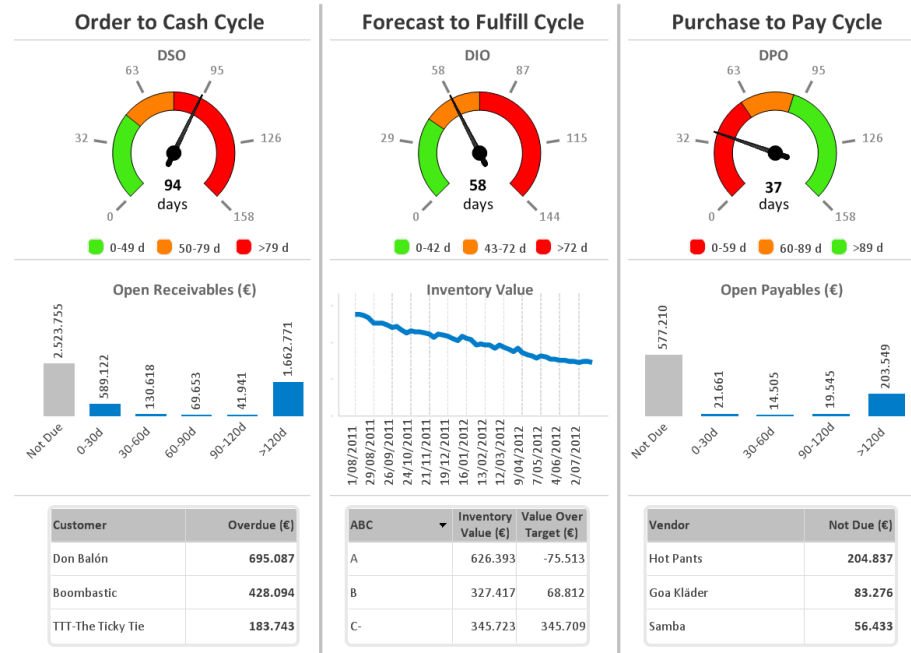


Dashboards

Provide a comprehensive visual view of corporate performance from multiple business areas

measures, trends, and exceptions

Allows executives to explore the situation in seconds



Pie Chart

Show percentages or proportional share

Used for component comparison

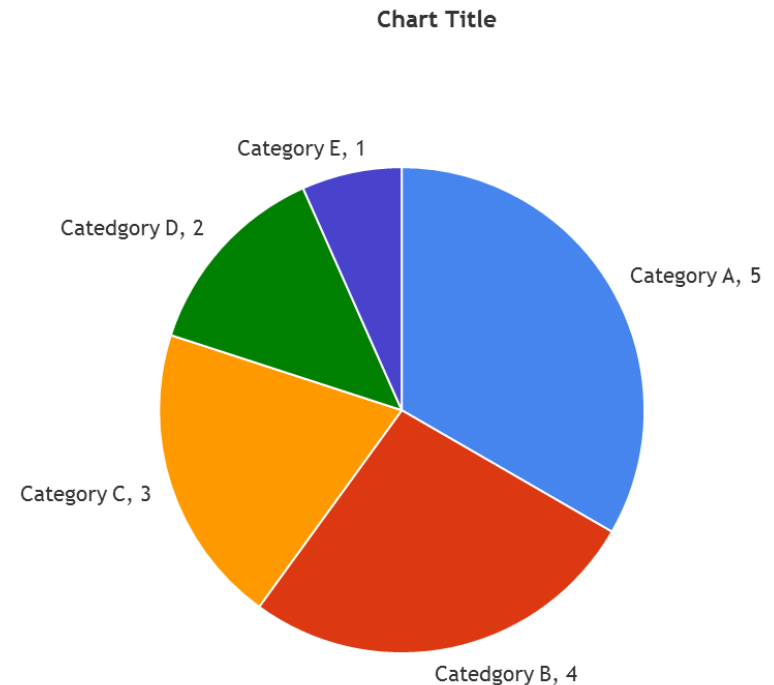
A circular chart divided into sectors, illustrating proportions. The arc

length of each sector (and consequently its angle and area)

is proportional to the value it represents

To represent the different parts of a whole
or the percentages of a total

Name	Value
Category A	5
Category B	4
Category C	3
Category D	2
Category E	1



Pie Chart:

Use a pie chart when you are making a point about the size or percentage of each component compared to a whole.

Your eye is used to measuring in a clock-wise motion, so you should position the most important segment against the 12 o'clock line.

To make the most of pie charts, do not have more than 6 slices.

Select the five most important components and make the rest into other.

Pie charts are the least practical of the chart forms and most often misused.

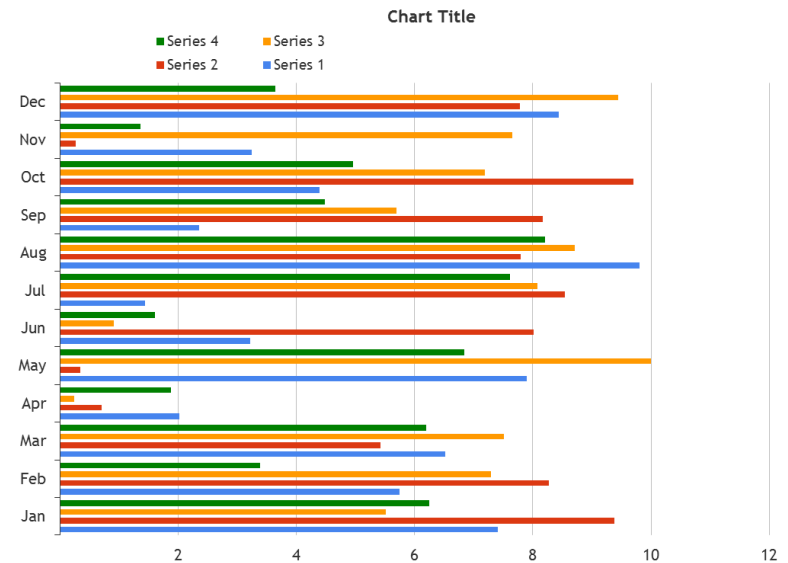
Bar chart

comparisons, categorical data

Used for item comparison

Uses rectangular bars with lengths proportional to the values they represent.

Often used to display and compare discrete data, or categorical data



Bar Chart:

When you compare different items, use a bar chart.

The vertical is used to label and measure different items

It is not a scale

You have some choices in how you arrange the bars or items.

You could arrange them in alphabetical order, or low to high or high to low.

Think about what order best stresses the point you want to make

Use can also use color to reinforce the what the data is telling you or the title of the chart.

Zelazny also suggests that the space between the bars be smaller than the size of the bars.

You can add a scale at the top or numbers at the end of the bars

but not both because it adds clutter.

Use the scale if all you want is a quick study of relationships, but use the numbers if they are important to your message.

Sometimes, you might want to use the scale and the one number that needs emphasis.

Types

Six variations:

Deviation

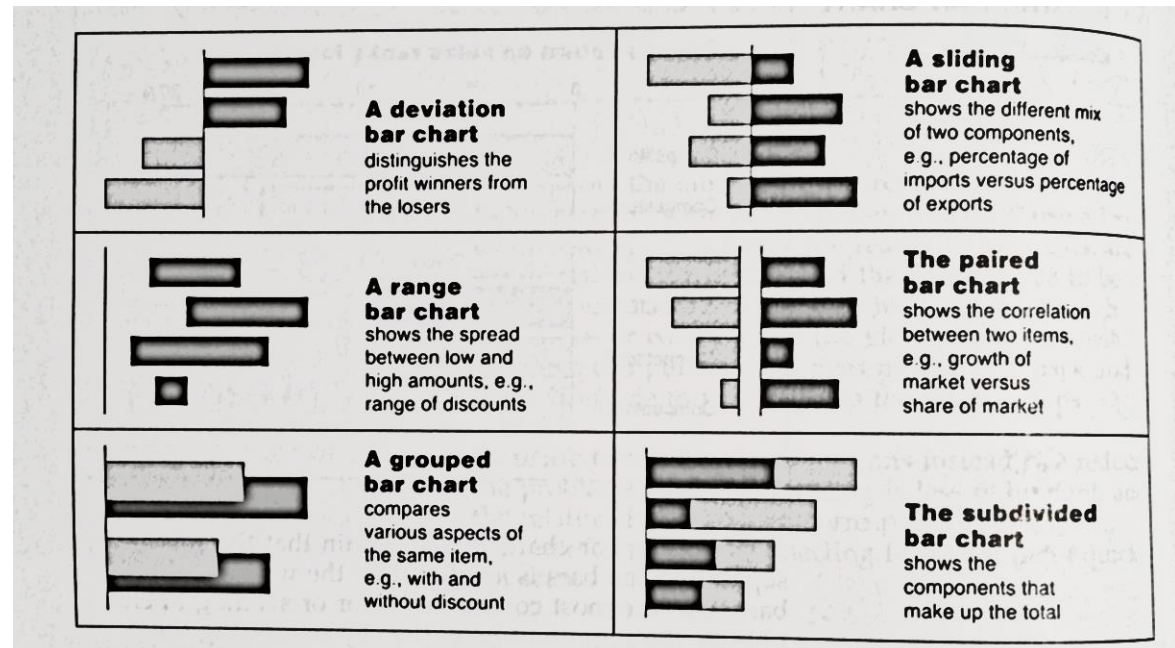
Sliding

Range

Paired

Grouped

Subdivided



Column chart

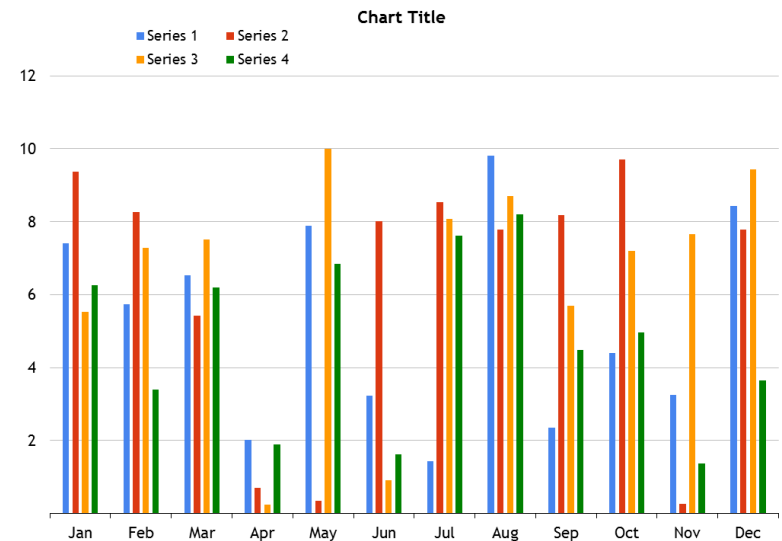
Best suited for time series
comparison

*If there are just a few values (up to
seven or eight)*

*Suggest fresh start between
columns*

Make space between columns
smaller than the width

Use color shading to emphasize
one point in time



Column chart – Variations

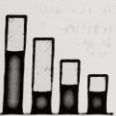
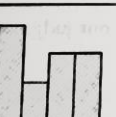
Deviation

Range

Grouped

Subdivided

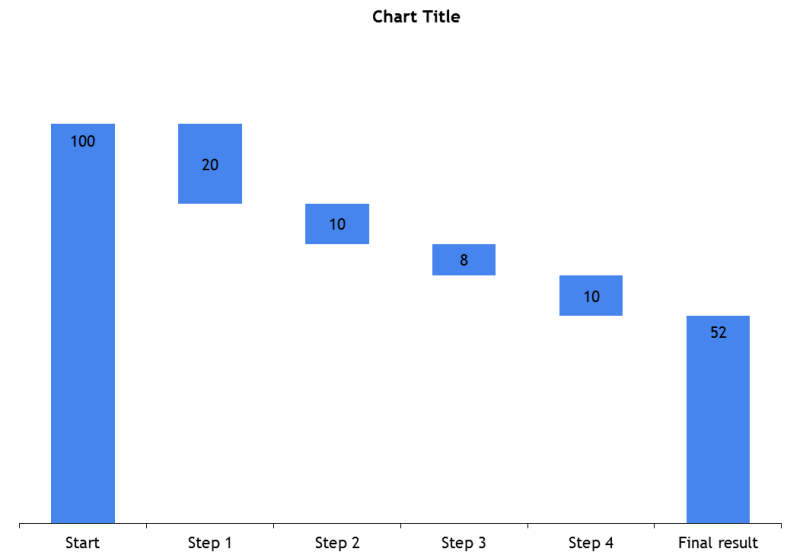
Step-column

	The deviation column chart distinguishes, say, the profit-making years from the losing ones.
	The range column chart shows the spread between low and high amounts, like the barometric performance of the stock market
	The grouped column chart, with columns either butting against one another or overlapping, compares two items at each point in time and shows how the relationships change over time, e.g., dollars with inflation and discounted for inflation.
	A subdivided column chart shows how the components contributing to the total vary over time, e.g., salary plus fringe benefits adding to total compensation.
	A step-column chart leaves no space between the columns and is best used to show data that change abruptly at irregular intervals, such as personnel ceilings or capacity utilization.

Column chart – Variations

Waterfall

Shows the change between points



Bar chart and histogram

Bar chart

Distinct values comparison

X-scale discrete

Histogram

Comparing distributions

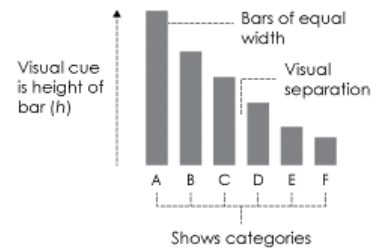
Related values

X-scale usually continuous

Though discretised

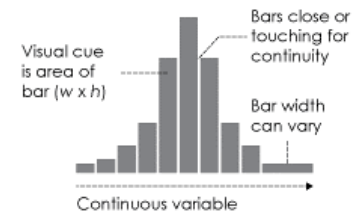
BAR CHART

Useful for comparing distinct values



HISTOGRAM

Useful for comparing distributions



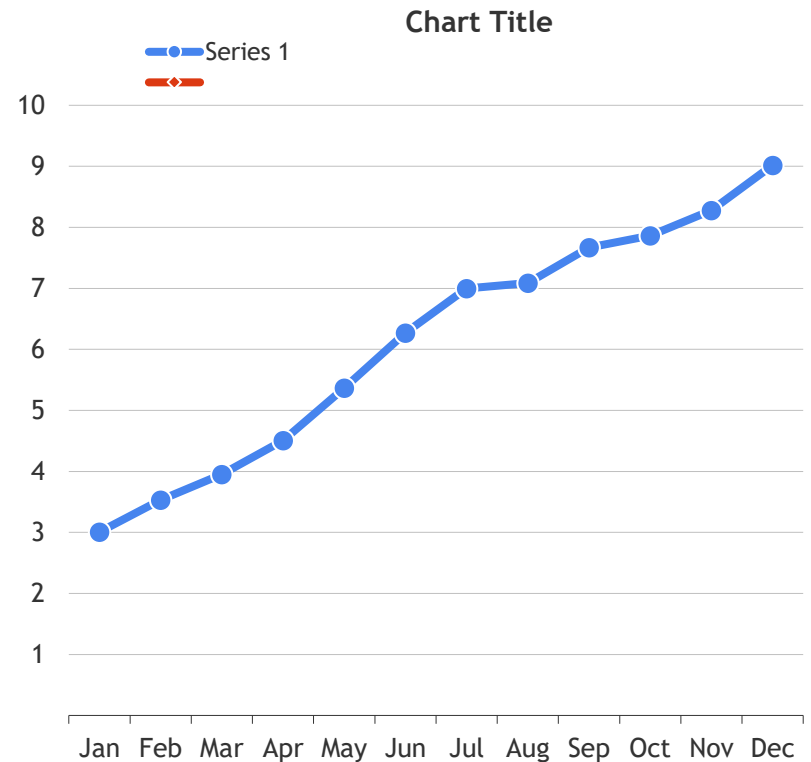
Line chart -

Shows trends in data at equal intervals

Best suited for time series comparison

If there are more values

Suggest flow between points

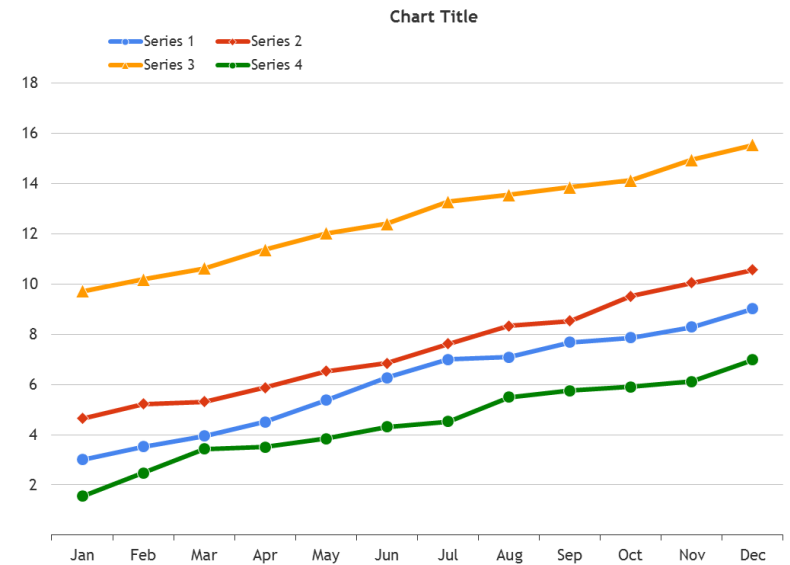
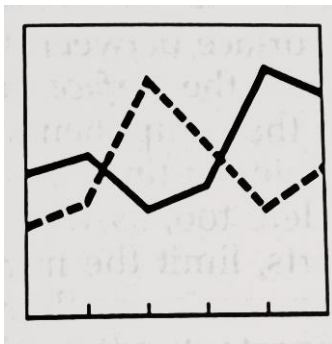


Line chart – variations

Grouped line chart

Compares performance of two or more items

How many lines ?

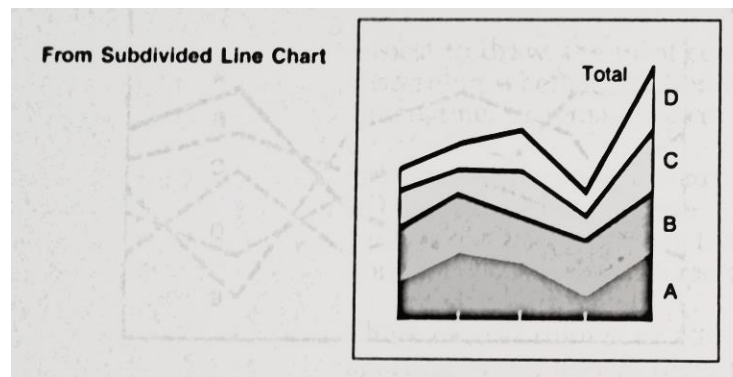


Line chart – variations

Surface chart

Apply shading to the surface between the line and the baseline

Subdivided surface chart:



XY chart/scatter (Dot chart)

illustrate relationship between two variables

Example:

16 transactions in terms of the size of the discount and the number of units sold

Expected – the larger the discount the greater the incentive to purchase more units

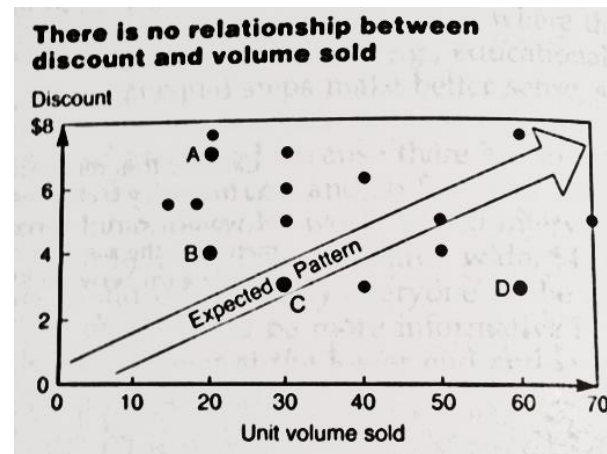
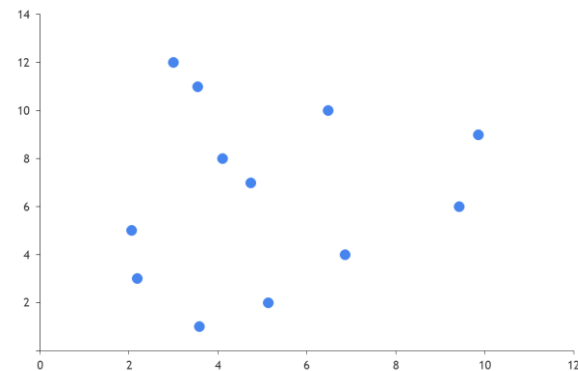


Chart Title



Thus, a scatter chart is used when:

- There are a large number of discrete data points.

- There are two variables that are inter-dependent.

- If there are two or more sets of data (multi-series scatter) and you want to show similarity in their variations with respect to a single parameter.

The common parameter is usually plotted along the horizontal axis.

XY chart/scatter (Dot chart)

It's often a good idea to demonstrate the expected pattern

Not to mix with line of best fit

Do not identify the dots – just causes confusion

Better use paired bar chart

Grouped dot chart

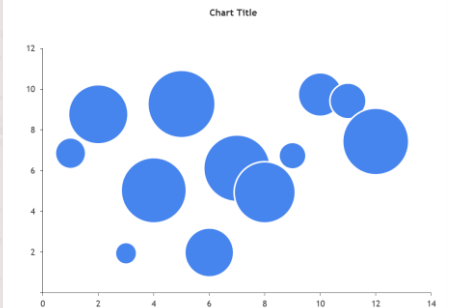
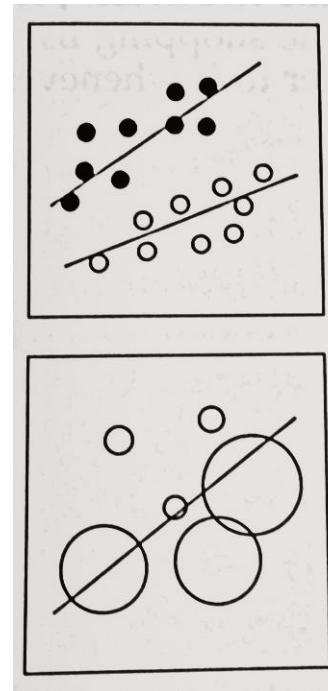
Groups of dots are identified (solid, open)

Top chart

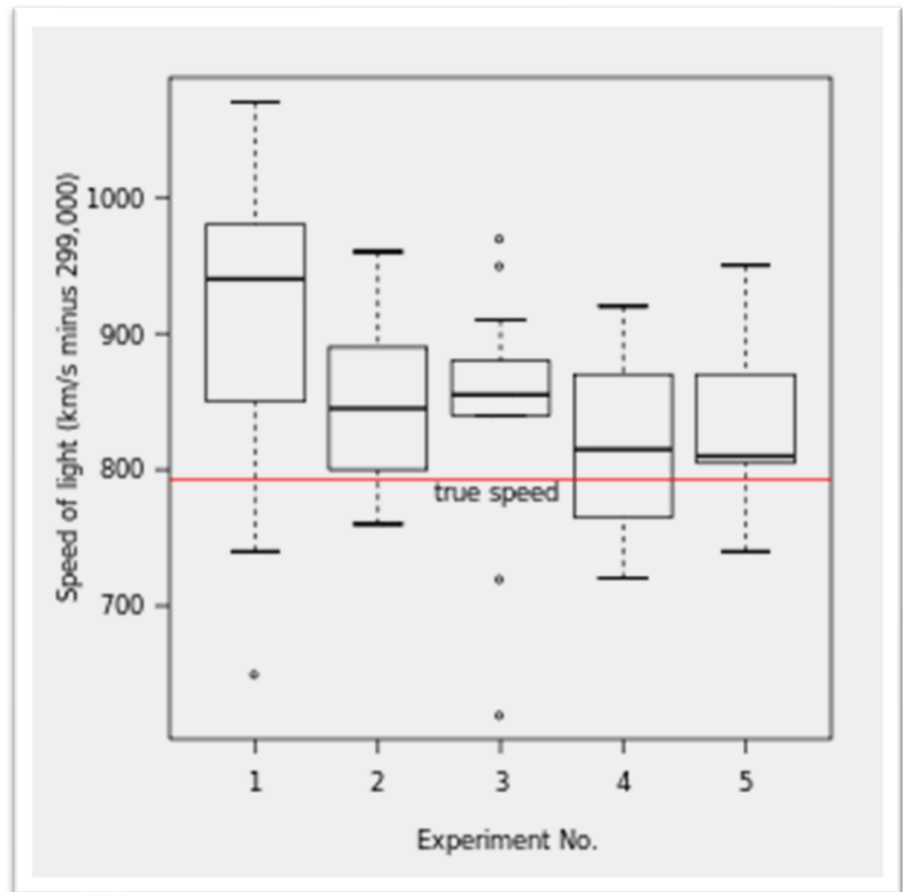
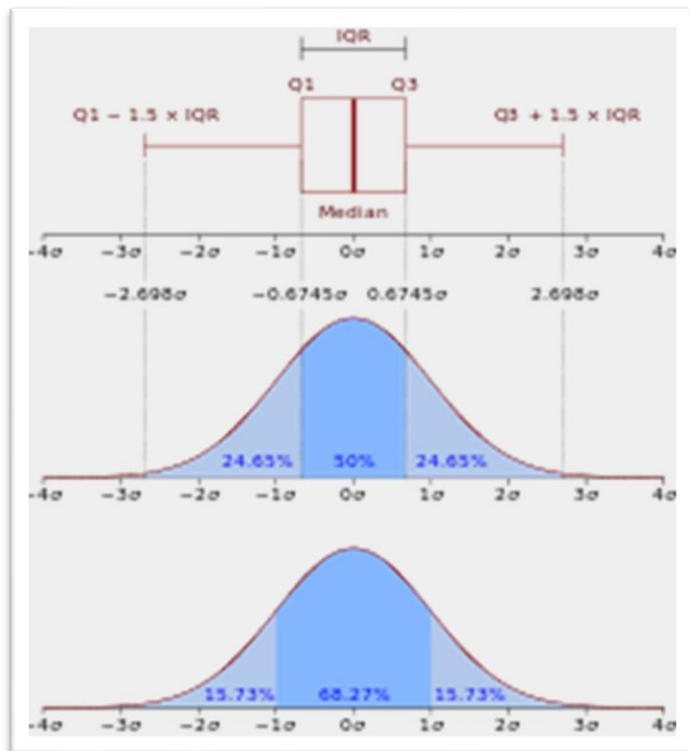
Bubble chart

Different bubble sizes

Bottom chart



Boxplot



Format

Finally, Format Your Chart

Once you've selected the right type of chart for your data

make sure you don't do your data a disservice by forgetting some basic design tips.

Kill the grid lines

unless they're absolutely necessary, or at least make them subtle so they don't distract from the information you're trying to present.

Make sure your chart is centered on the data you want to present,

your axes are clearly labeled, and your axes have units on them where necessary,

so no one has to guess or infer what you're trying to say.

Remember,

your goal is that anyone can pick up your chart, whether you're there to talk to it or not, and understand what information the data is trying to communicate.

When using graphics

- Ensure graphic has a heading
- Label the components of your graphic
- Indicate data source
- Use the graphic to work for you – sort the data, or add labels, use colour

Points to remember

- Graphics should stimulate thinking
- Use a graphic if it can't be put in a table or adds more value as a graphic
- Graphics are useful for portraying trends, relationships and comparisons
- Maximise the data to ink ratio – do not waste ink
- Minimise chart or graphic junk
- Never cut the y-axis of a graph

Six Steps to Great Graphs

Step 1: Which Chart is Best?

Step 2: Use Color To Emphasize

Step 3: Delete Data You Don't Need

Step 4: Directly Label

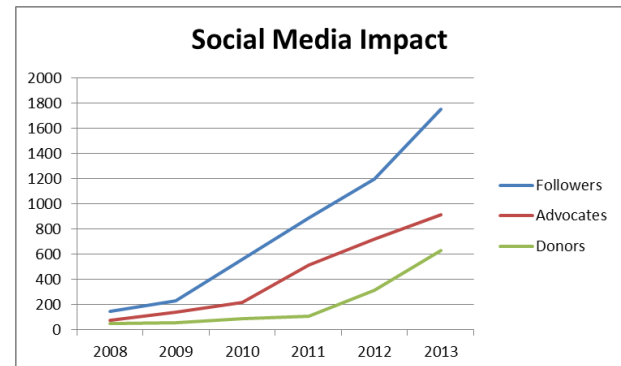
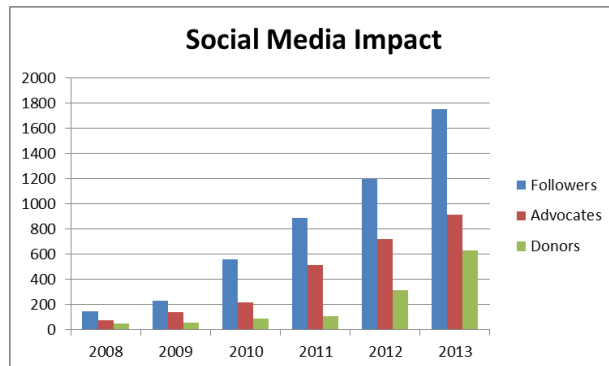
Step 5: Annotate

Step 1: Which Chart is Best?

People are better at judging length, so bar charts are a decent option also useful for comparisons.

When trying to look at change over time

line graphs are a more appropriate chart type than the bar graph

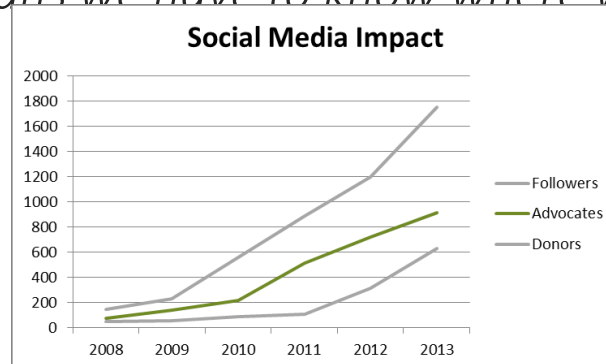


Step 2: Use Color to Emphasize

it can be difficult to tell the graph's main point, without some serious cognitive effort.

Changing the color can help bring attention where we want it.

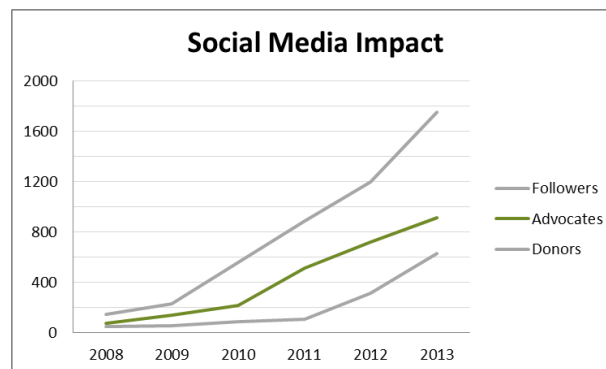
Of course, that means we have to know where we want it.



Step 3: Delete What You Don't Need

Little things in this chart still make it feel cluttered and distract from the data.

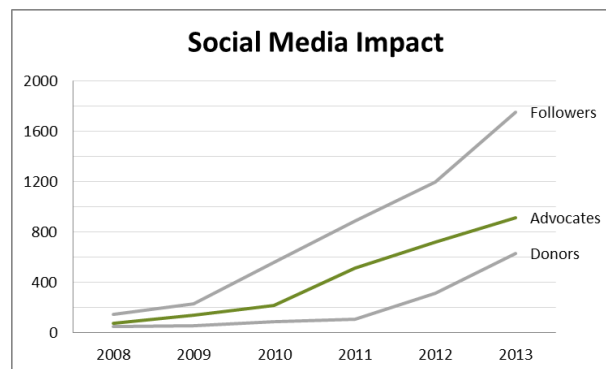
Most of the time, we can do without the tick marks along each axis.



Step 4: Directly Label

Our brains make better sense of the data when we replace the legend with direct line labels.

This way, interpretation of which line stands for what is very obvious.

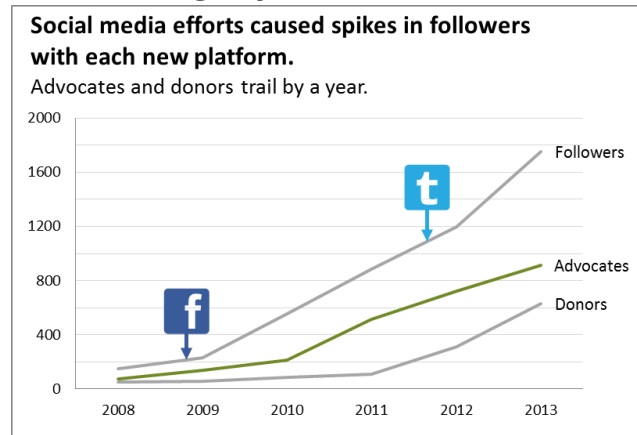


Step 5: Annotate

Let's get back to the original reason we visualize data – we're looking for a pattern.

We graph our social media impacts because we need to see how launching new platforms has influenced our clients.

So now that we have taken lots of things out of the graph, *let's add back in some thoughtful annotation to help make the patterns clear.*



Some tips

Identify the message

Pick the right chart type for the message and numerical data

Add enough text, avoid abbreviations!

Avoid 3D in 2-dimensional visualizations!

Bar-charts: always with zero-baseline

Line-charts: sometimes better w.o zero-baseline

No axis-breaks – they are never intuitive

Show all data, and show also MISSING data

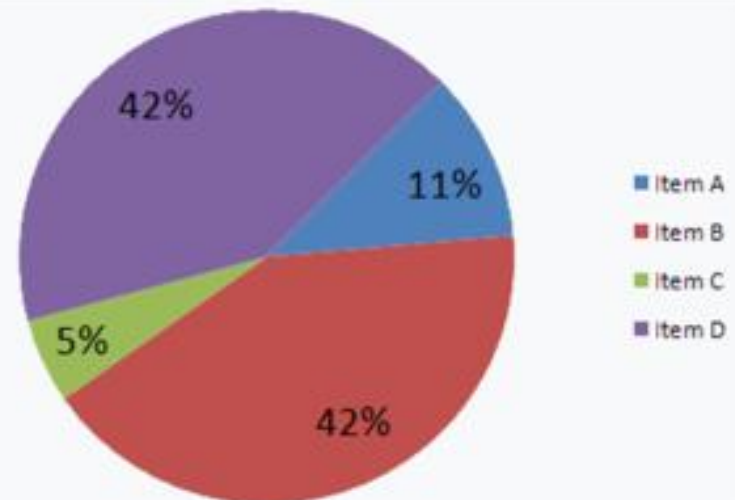
Make colors accessible to all audiences!

Comparison of pie charts

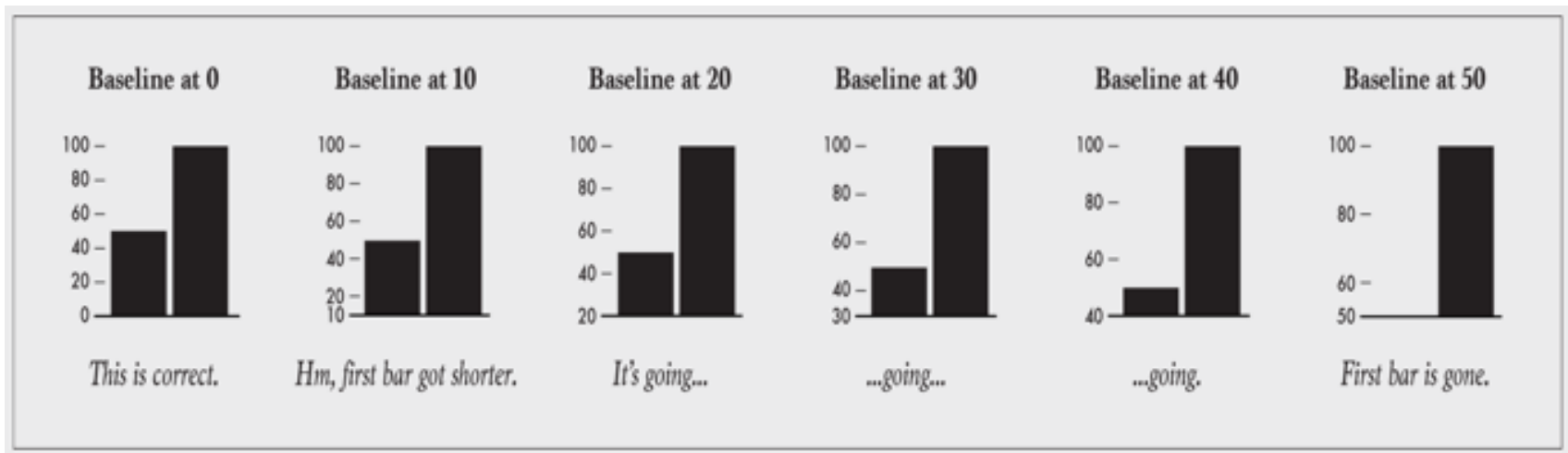
Misleading pie chart



Regular pie chart

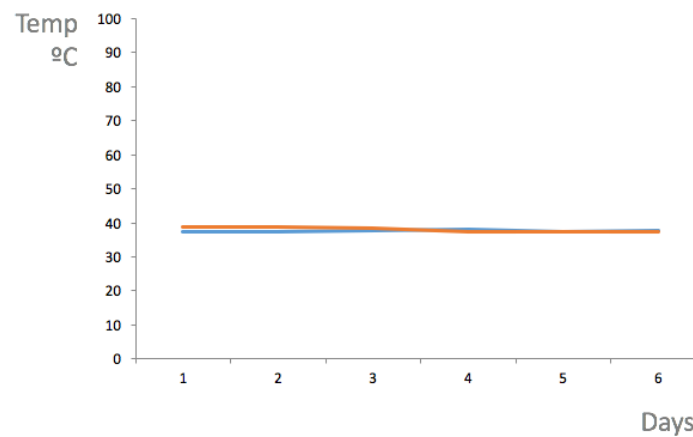


In the misleading pie chart, Item C appears to be at least as large as Item A, whereas in actuality, it is less than half as large.



Patient fever chart

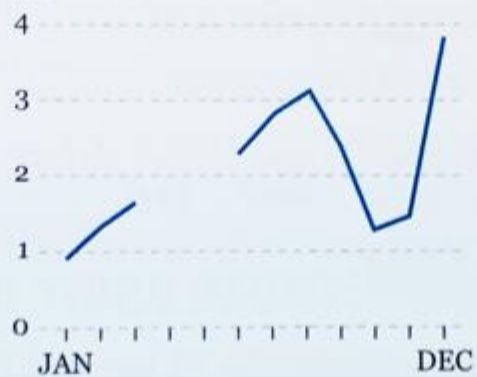
...nobody has fever here!

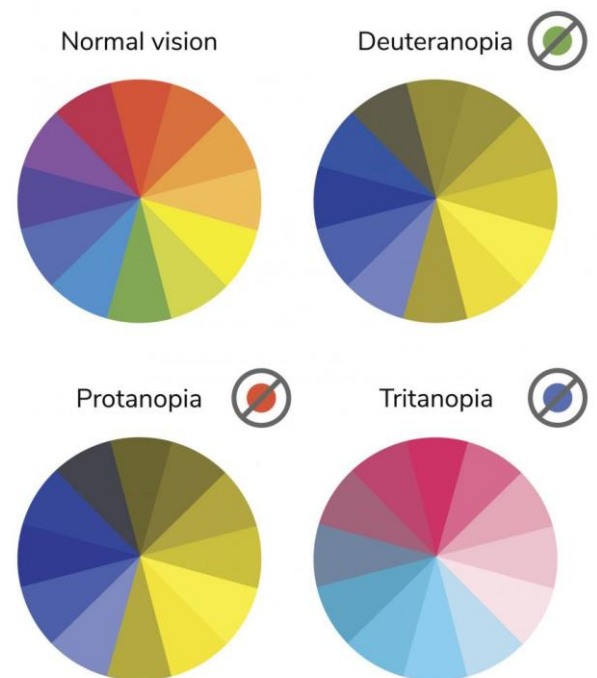
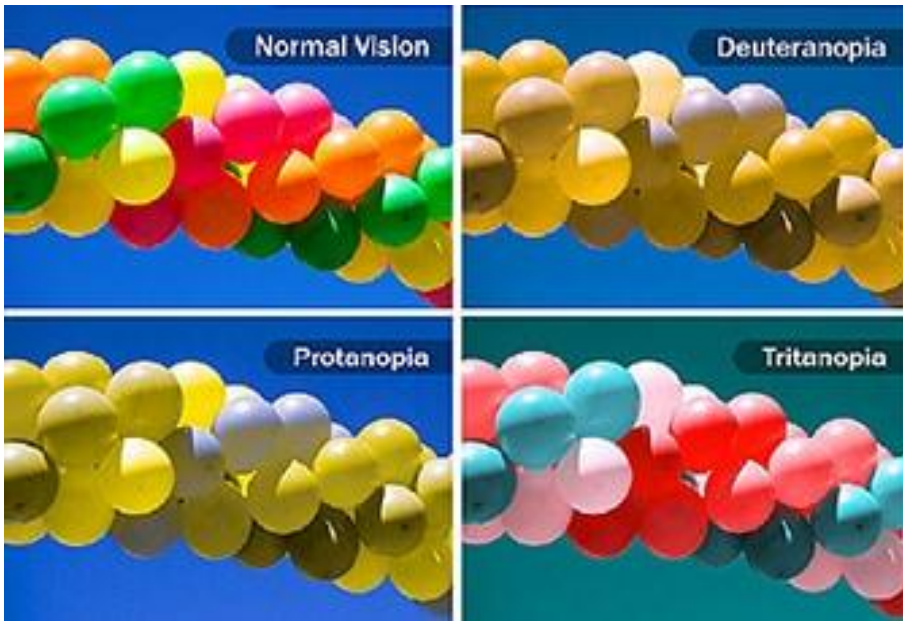


Misleading choice



Better choices





Choosing good colors for your charts is hard.

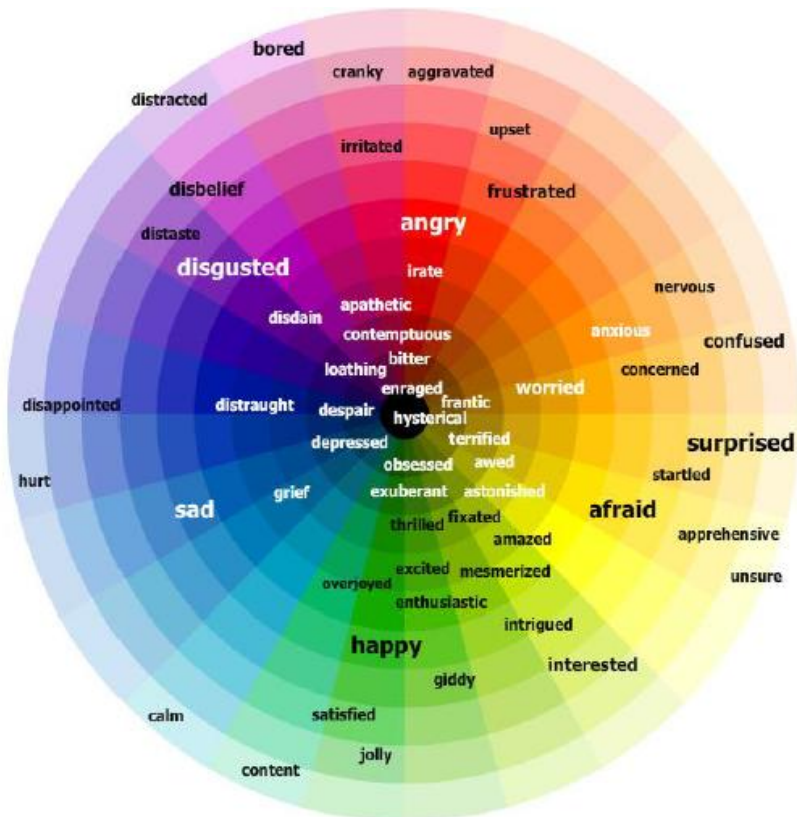
Sequential values:
1 Color **vary saturation**



2 Colors for **diverging** values

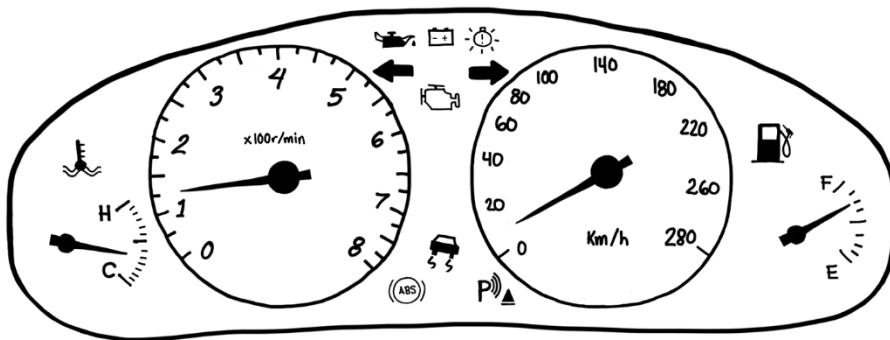


Categorical data
vary the hue





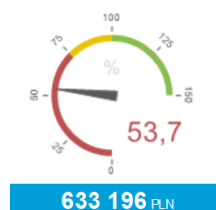
Dashboard



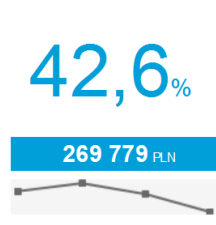
Forte Business Intelligence

Sprzedaż

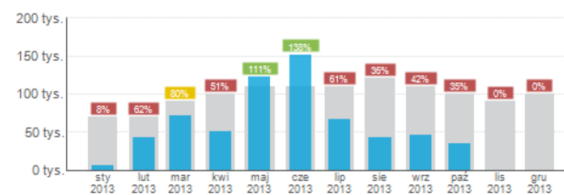
PLANY



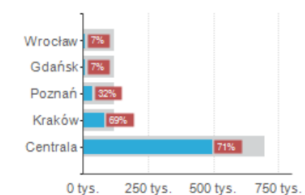
MARŻA



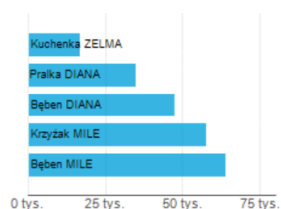
Realizacja planów w ujęciu miesięcznym



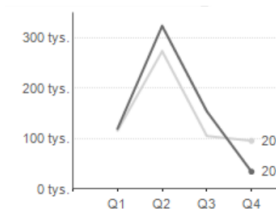
Realizacja planów wg działów Top 5



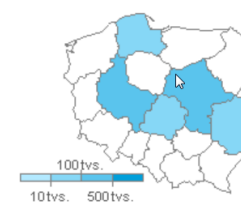
Towary z najwyższą marżą Top 5



Sezonowość sprzedaży

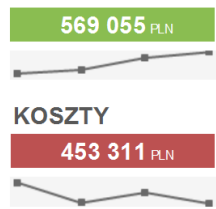


Sprzedaż w ujęciu geograficznym

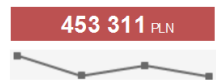


Finanse

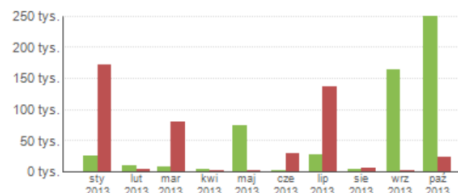
PRZYCHODY



KOSZTY



Przychody i Koszty



Analiza zysku

	2012	2013
Ze sprzedaży:	88 527	80 075
Z działalności operacyjnej:	104 460	124 970
Z działalności gospodarczej:	104 460	115 744
Zysk brutto:	113 031	115 744
Zysk netto:	113 031	115 744

How Are We Doing?

Revenue

\$1,527,881

MRR

\$617,621

Operations Cost

\$2,389,320

Total Subscriptions

8,519

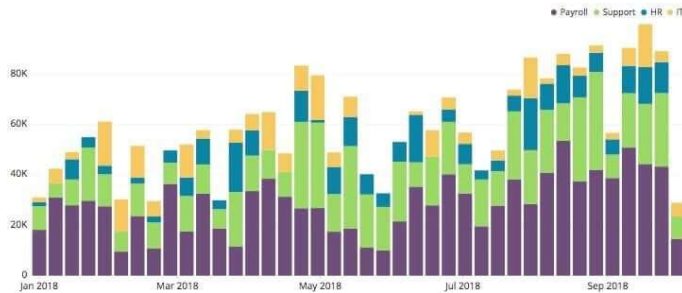
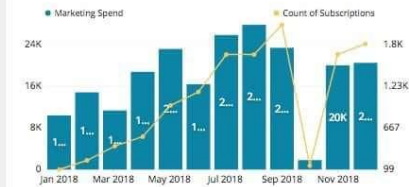
Marketing Spend

\$173,875

MRR Trend



Spend vs. Customer Acquisition with Forecast



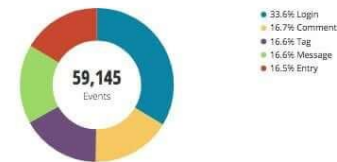
Count of Users

34,696

User Count Trend (This Year)



Activity



Dashboard

Combining graphs and charts

Positioning is very important

Graphs with the same set of data or similar topics should be placed together.

Dashboards should not require users to jump back and forth from graphs to graphs.

Graphs should be positioned based on their visual hierarchy.

Graphs displaying overview data should have a position at the top.

Details are normally presented in the area below.

Good navigation will allow users to construct a story from graphs provided in a dashboard.





Process

What business process will you represent with the model/reports?



Outcomes

What does the business want to achieve if the process is successful?



Users

Who will use the model/reports and what their role is in the process?



Decisions and actions

What do users do to try and achieve their outcomes?



Problems and questions

What data challenges do users face to make decisions or take actions?



Data

What information do users need to address their challenges?



Focus on specific questions

Avoid creating a report that tries to show everything.



Go from top-left to bottom-right

Place the most important information in the top-left of the report.



Use color to steer attention

Make it quick and convenient for users to get what they need.



Limit ink and information

Avoid overwhelming users with too much to see and think about.



Keep it simple

Choose the right charts for the data, users, and questions.



Make it convenient

Help users spend as little time on the report as possible.

Dashboard design principles also suggest giving users immediate access to the most important data.

the “five-second rule”

all relevant data should be reachable within five seconds after the dashboard loads

work with users to determine which data is most relevant to their business objectives

display this data in a prominent spot on the dashboard

display the remaining information in a logical order - the typical layout of a dashboard:

starts with the most significant data at the top of the screen.

the middle section is reserved for trends

granular details are placed at the bottom

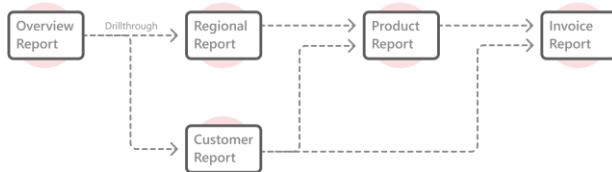


Detail

A flow that depicts how users navigate reports to answer their questions.



Reports or report pages in a single workspace or app



1. INDICATORS

2. TRENDS

3. DETAILS



How Are We Doing?

Revenue

\$1, MOST IMPORTANT

MRR

Operations Cost

\$864,844

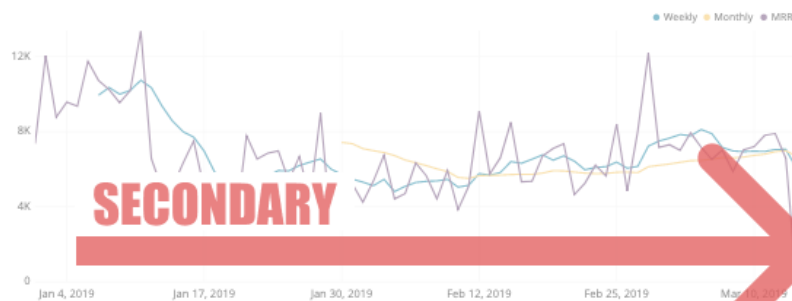
Total Subscriptions

8,519

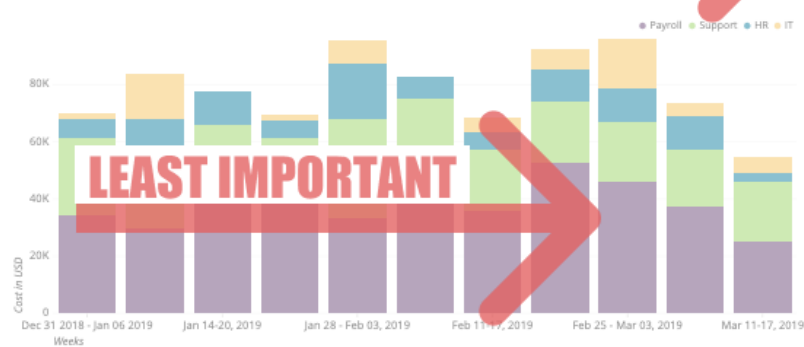
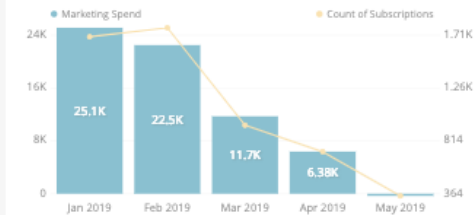
Marketing Spend

\$59,369

MRR Trend



Spend vs. Customer Acquisition with Forecast



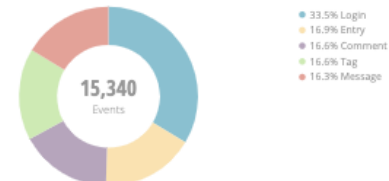
Count of Users

34,696

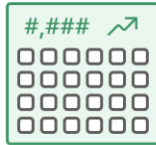
User Count Trend (This Year)



Activity



3-30-300



3 seconds:

Users should get an overview of the most important questions and areas.



30 seconds:

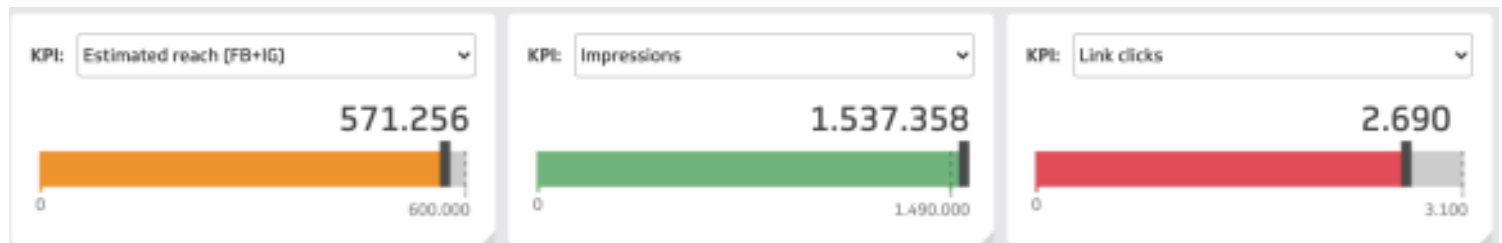
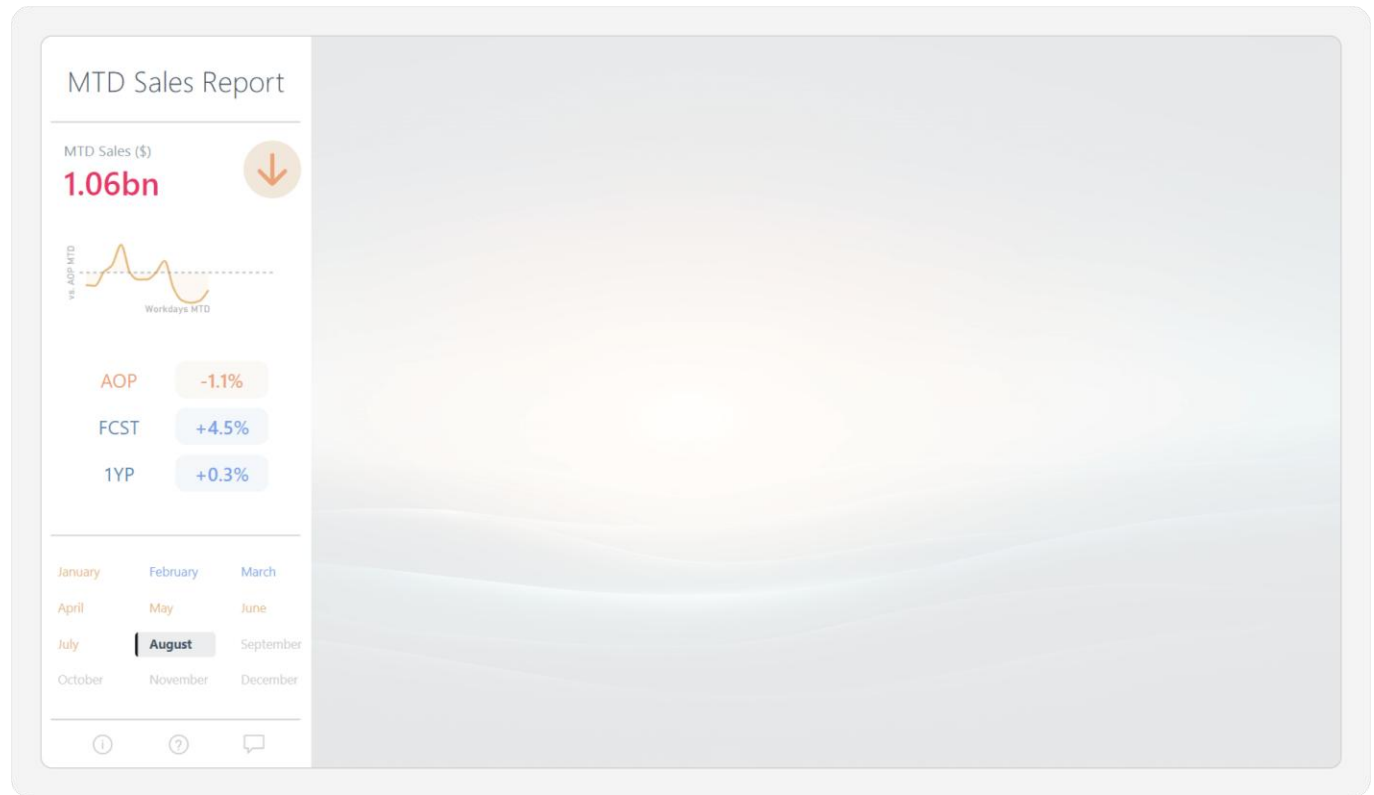
Users should filter and zoom to identify periods and categories to focus on.



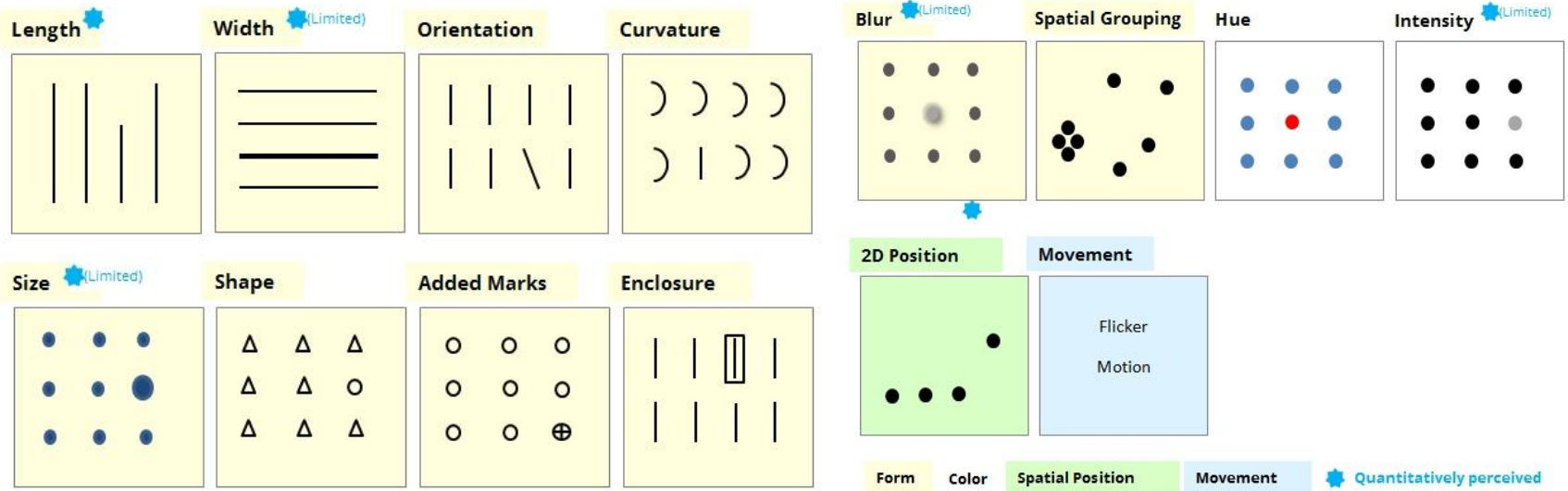
300 seconds:

Users should get details-on-demand to inform their decisions and actions.

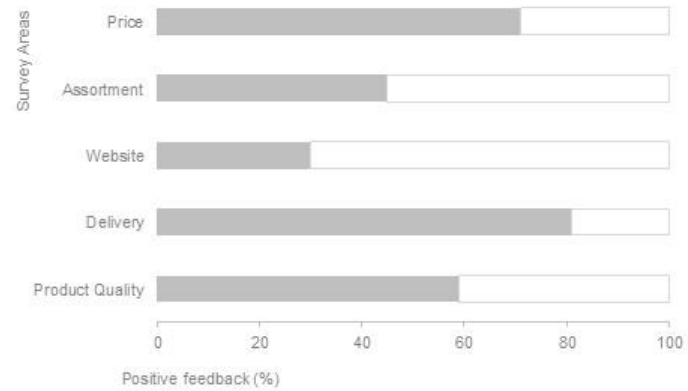
3



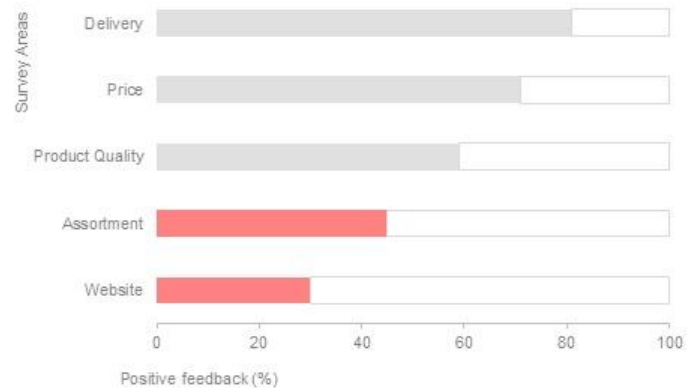
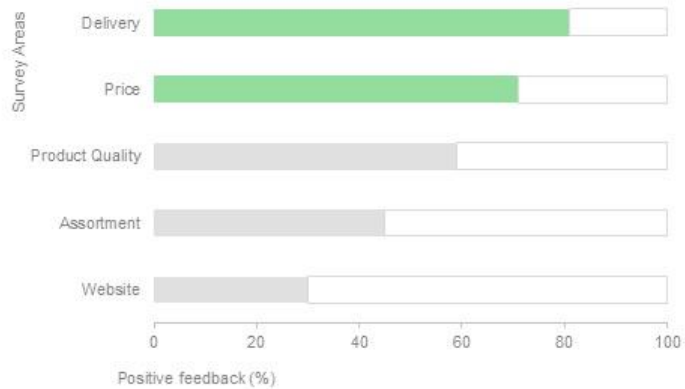
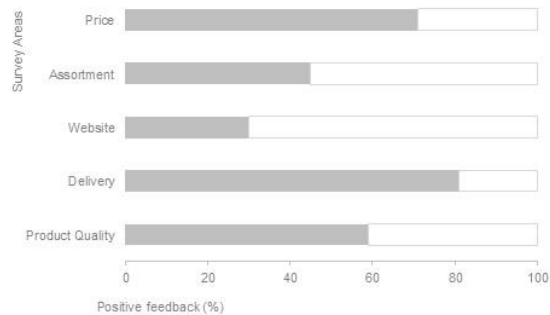
3



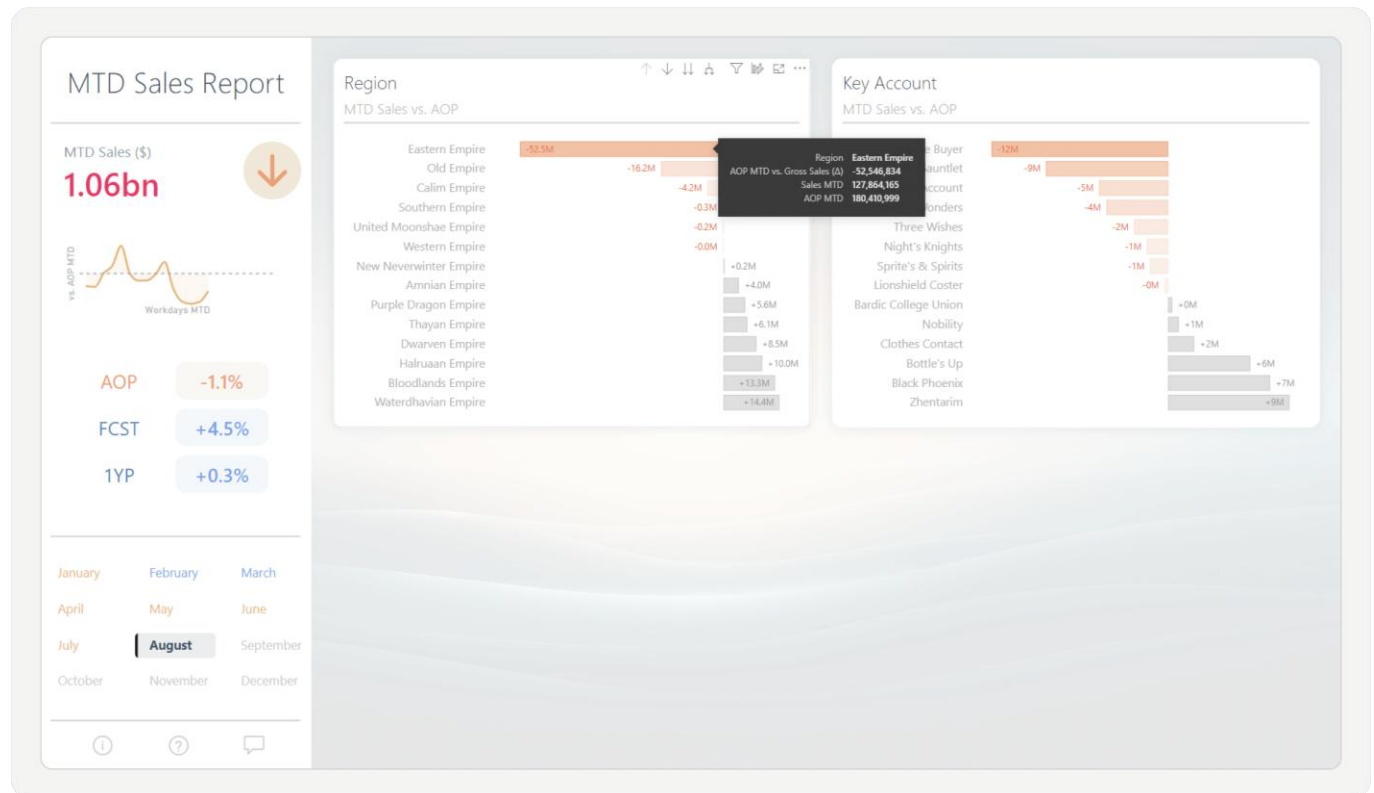
3



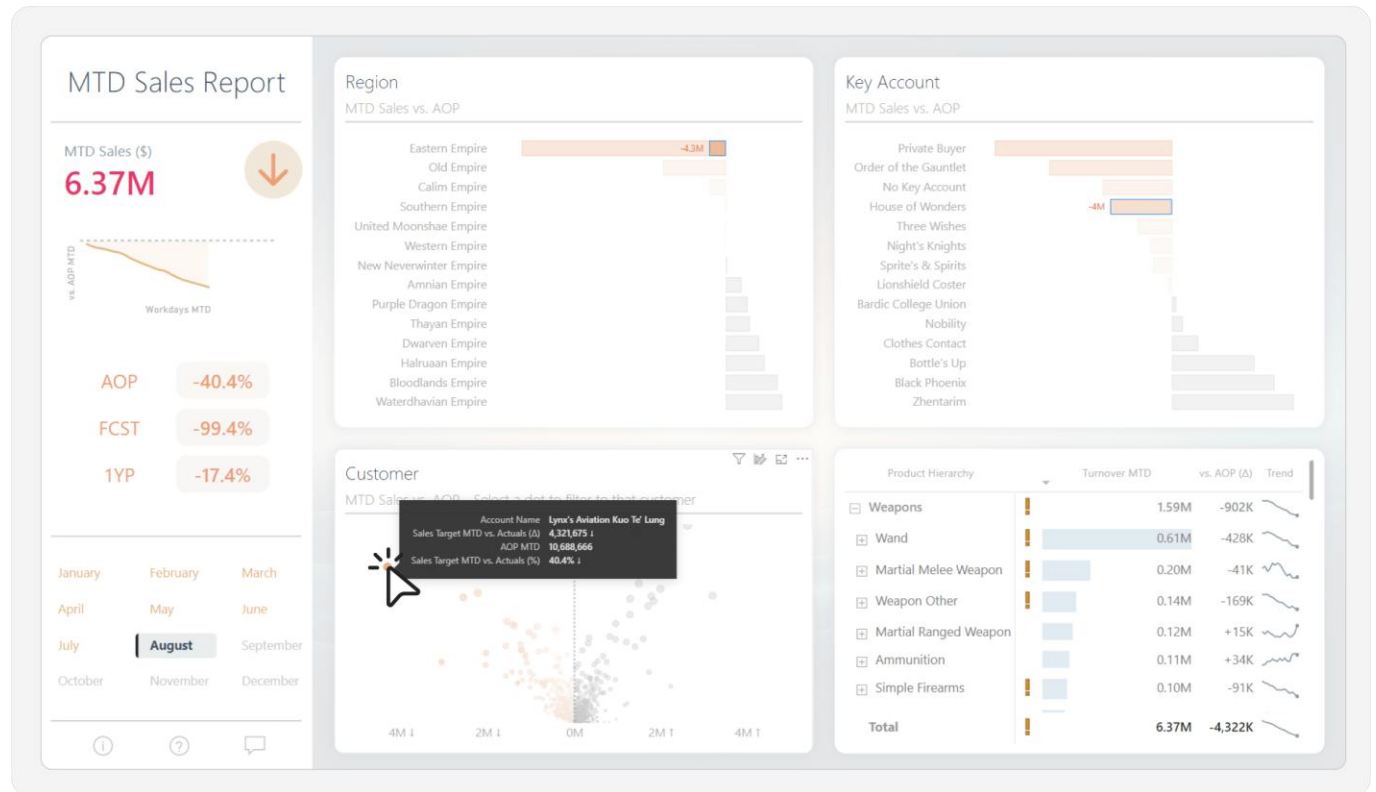
3



30



300



Facebook & Instagram Posts

Network: All Targeting: All

Ad ...	Post D...	Ad Previ...	Targeting	Post Link	Impres...	Reacti...	Comm...	Shares	Link CL...	Video ...	Estim...	Ad Re...	Ad Re...
f	Jun 16, 2				178,274	61	2	10	223	0	104,050	6,430	6,2%
f	Jun 16, 2				129,171	58	1	11	148	0	78,480	5,580	7,1%
f	Jun 16, 2				105,415	24	1	11	111	0	60,340	3,850	6,4%
f	Jun 16, 2				67,548	22	1	1	79	0	40,120	2,740	6,8%
f	Jun 16, 2				68,005	30	1	6	82	0	39,660	2,800	7,1%
f	Jun 16, 2				79,327	18	0	11	99	0	47,220	2,720	5,8%



Doesn't follow the 3-30-300 rule

MTD Sales Report

January
MTD Sales: 1.53 bn

February
MTD Sales: 1.33 bn

March
MTD Sales: 1.61 bn

April
MTD Sales: 1.64 bn

May
MTD Sales: 1.59 bn

June
MTD Sales: 1.64 bn

July
MTD Sales: 1.53 bn

August
MTD Sales: 1.06 bn

September
MTD Sales: --

October
MTD Sales: --

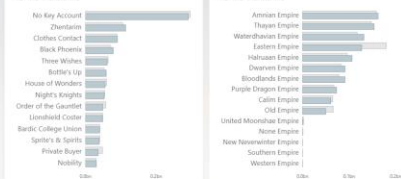
November
MTD Sales: --

December
MTD Sales: --

Customer Sales MTD

by Key Account

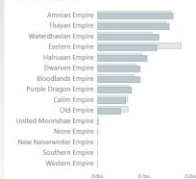
1 AOP MTD 8 Turnover MTD



Regional Sales MTD

by Commercial Region

1 AOP MTD 8 Turnover MTD



MTD Sales (\$)

AOP MTD 1.07bn -1.0%

FCST MTD 1.07bn -0.9%

1YR MTD 1.06bn -0.9%

MTD Sales (\$)

1YR MTD 7.83M

MTD Sales (\$)

1YR MTD 229K

MTD Sales (\$)

1YR MTD 506M

MTD Sales (\$)

1YR MTD 7.57M

MTD Sales (\$)

1YR MTD 176K

MTD Sales (\$)

1YR MTD 68.6%

MTD Sales (\$)

1YR MTD 51.2%

MTD Sales (\$)

1YR MTD 64.9%

Sales Trend MTD

vs. Annual Operating Plan (AOP) MTD



Follows the 3-30-300 rule

MTD Sales Report

MTD Sales (\$)

1.06bn

AOP MTD

FCST

1YR

AOP -1.1%

FCST +4.5%

1YR +0.3%

January

February

March

April

May

June

July

August

September

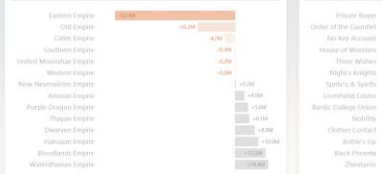
October

November

December

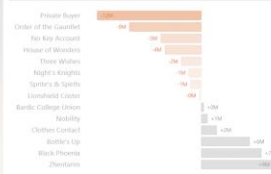
Region

MTD Sales vs. AOP



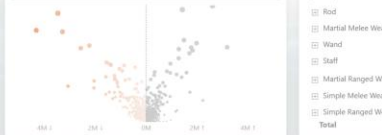
Key Account

MTD Sales vs. AOP



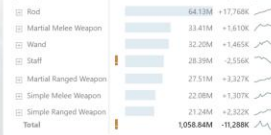
Customer

MTD Sales vs. AOP - Select a dot to filter to that customer



Product Hierarchy

Turnover MTD vs. AOP (Δ) Trend



Doesn't follow the 3-30-300 rule

MTD Sales Report

Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec

Key Account

	Turnover MTD	AOP MTD	vs. AOP (%)	vs. AOP (Δ)	Trend	1YR MTD	vs. 1YR (%)	vs. 1YR (Δ)	2YR MTD	vs. 2YR (%)	vs. 2YR (Δ)
No Key Account	292.05M	296.90M	-1.6%	-4.85M		310.72M	-6.0%	-18.67M	269.28M	-6.0%	-18.67M
Armour	73.61M	81.59M	-9.8%	-7.98M		85.53M	-13.9%	-11.92M	77.98M	-13.9%	-11.92M
Weapons	69.39M	63.94M	+8.5%	+5.45M		66.73M	+4.0%	+2.66M	60.04M	+4.0%	+2.66M
Conjuration	34.35M	37.45M	-8.3%	-3.10M		38.85M	-11.6%	-4.50M	29.68M	-11.6%	-4.50M
Mobility	16.20M	22.16M	-26.9%	-5.96M		22.99M	-29.5%	-6.79M	20.18M	-29.5%	-6.79M
Adventuring	23.83M	21.54M	+10.7%	+2.29M		25.41M	-6.2%	-1.58M	21.03M	-6.2%	-1.58M
Tools & Kits	24.60M	21.43M	+14.8%	+3.17M		23.05M	+6.7%	+1.55M	16.62M	+6.7%	+1.55M
Enchantment	20.12M	18.27M	+10.1%	+1.84M		19.34M	+4.0%	+0.78M	17.75M	+4.0%	+0.78M
Instruments	19.15M	17.50M	+9.4%	+1.65M		18.17M	+5.4%	+0.98M	14.56M	+5.4%	+0.98M
Jewelry	10.79M	13.02M	-17.1%	-2.23M		10.64M	+1.4%	+0.15M	11.45M	+1.4%	+0.15M
Zhentarim	114.60M	106.08M	+8.0%	+8.52M		103.92M	+10.3%	+10.68M	93.10M	+10.3%	+10.68M
Weapons	41.72M	37.25M	+12.0%	+4.46M		35.63M	+17.1%	+6.09M	33.05M	+17.1%	+6.09M
Armour	24.50M	23.78M	+3.0%	+0.71M		23.74M	+3.2%	+0.75M	22.96M	+3.2%	+0.75M
Adventuring	12.41M	9.65M	+28.5%	+2.75M		10.72M	+15.7%	+1.68M	9.15M	+15.7%	+1.68M
Instruments	8.68M	8.14M	+6.6%	+0.54M		7.80M	+11.2%	+0.88M	6.65M	+11.2%	+0.88M
Tools & Kits	8.36M	7.50M	+11.5%	+0.86M		7.54M	+10.8%	+0.81M	5.30M	+10.8%	+0.81M
Enchantment	6.65M	6.08M	+9.3%	+0.56M		6.29M	+5.6%	+0.35M	4.16M	+5.6%	+0.35M
Mobility	5.56M	5.66M	-1.8%	-0.10M		5.55M	+0.2%	+0.01M	5.57M	+0.2%	+0.01M
Adventuring	4.10M	4.14M	-1.0%	-0.04M		4.01M	+2.3%	+0.09M	4.11M	+2.3%	+0.09M
Jewelry	2.64M	3.87M	-31.8%	-1.23M		2.64M	+0.1%	+0.00M	2.15M	+0.1%	+0.00M
Clothes Contact	91.67M	89.83M	+2.0%	+1.84M		91.97M	-0.3%	-0.30M	87.30M	-0.3%	-0.30M
Armour	29.62M	30.29M	-2.2%	-0.67M		31.15M	-4.9%	-1.53M	32.17M	-4.9%	-1.53M
Total	1,058.84M	1,070.13M	-1.1%	-11.29M		1,056.00M	+0.3%	+2.85M	960.79M	+0.3%	+2.85M



Follows the 3-30-300 rule

MTD Sales Report

MTD Sales (\$)

1.06bn

AOP MTD

FCST

1YR

AOP -1.1%

FCST +4.5%

1YR +0.3%

January

February

March

April

May

June

July

August

September

October

November

December

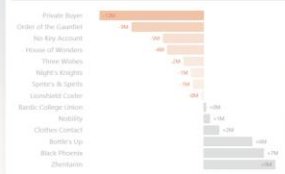
Region

MTD Sales vs. AOP



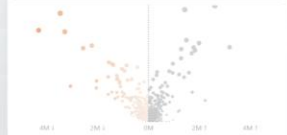
Key Account

MTD Sales vs. AOP



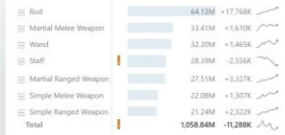
Customer

MTD Sales vs. AOP - Select a dot to filter to that customer



Product Hierarchy

Turnover MTD vs. AOP (Δ) Trend



Tell a story

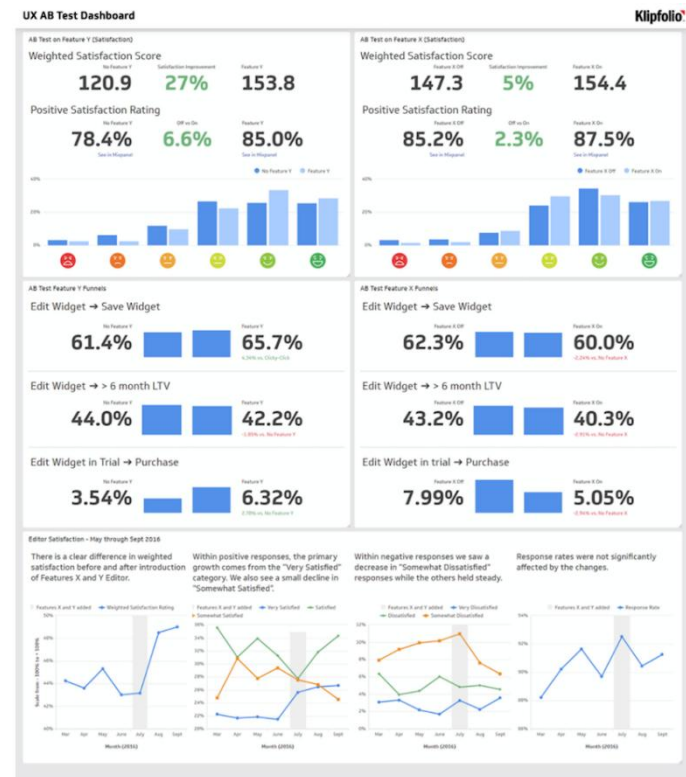
Remember:

Tell a story

Related data belongs together.

If one metric relates to another, save user time and group them together.

The most frequently checked information should be placed on the left side or above the related data.

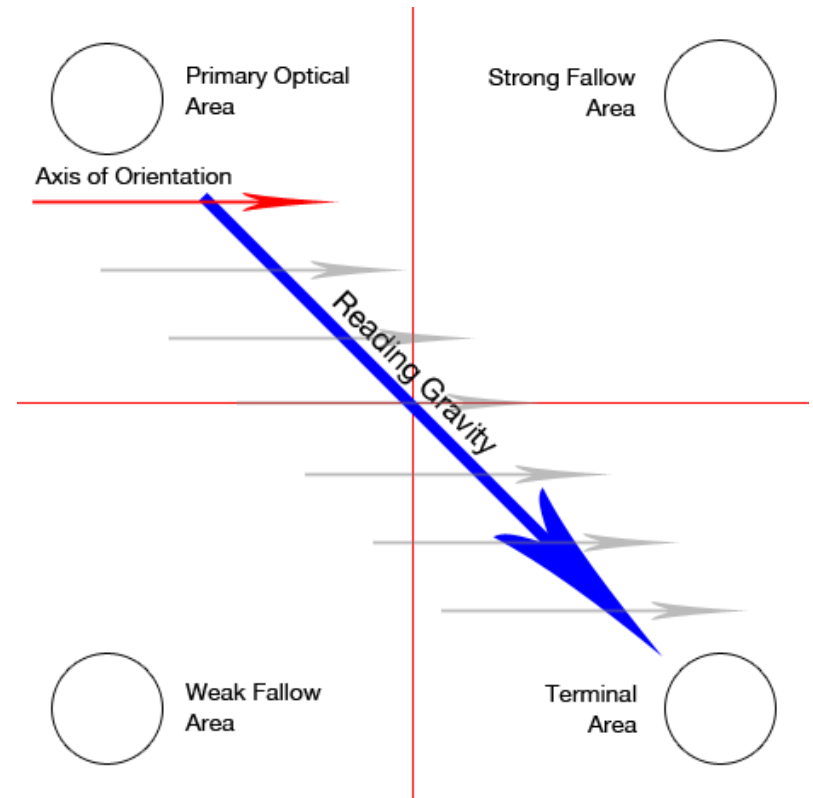


Z-pattern

Readers will start in the top/left, move horizontally to the top/right and then diagonally to the bottom/right before finishing with another horizontal movement to the bottom/right.

The z-pattern is good for simple designs with a few key elements that need to be seen.

Any storytelling aspect of the design would follow the path of the z



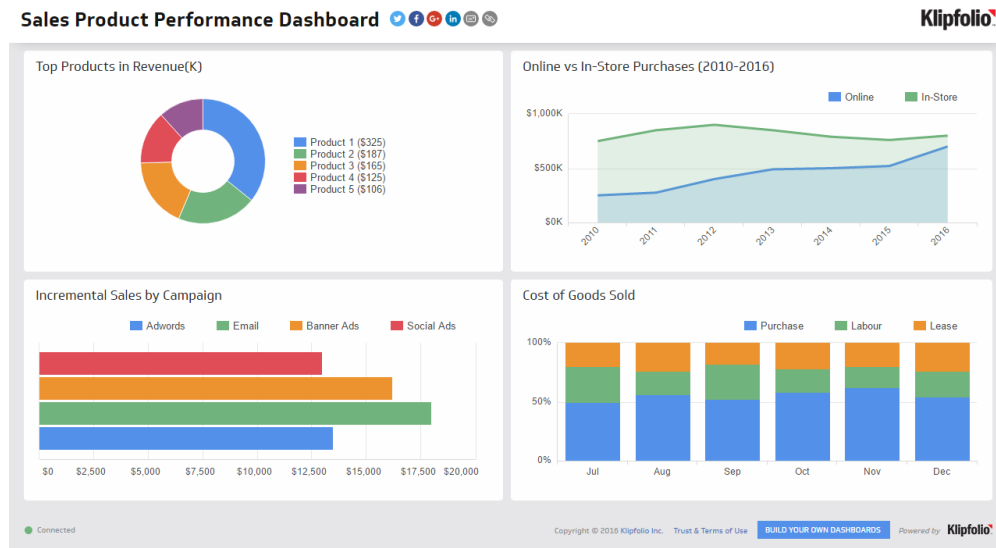
Finally, keep it simple and don't overcrowd your dashboard!

You'll want to stay between 7 – 9 widgets on each dashboard

More than this and you'll likely overwhelm viewers.

The Magic number 7 (plus or minus two) provides evidence for the capacity of short term memory.

Most adults can store between 5 and 9 items in their short-term memory.



An optimal dashboard is ACES:

Accurate

A dashboard lives and dies by the trust the viewers have in what they are seeing.

If the viewers doubt a dashboard's accuracy, it will not be used to make decisions.

Remember

Data quality

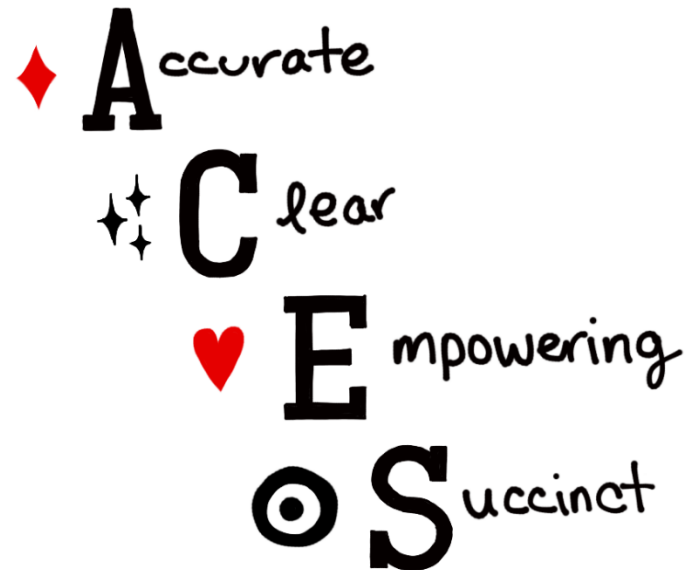
Metric comprehension

Visualization design

Clear

Empowering

Succinct



An optimal dashboard is ACES:

Accurate

Clear

Data must be displayed clearly

Fonts

Adequate to device and message

Colors

Adequate to message

Context

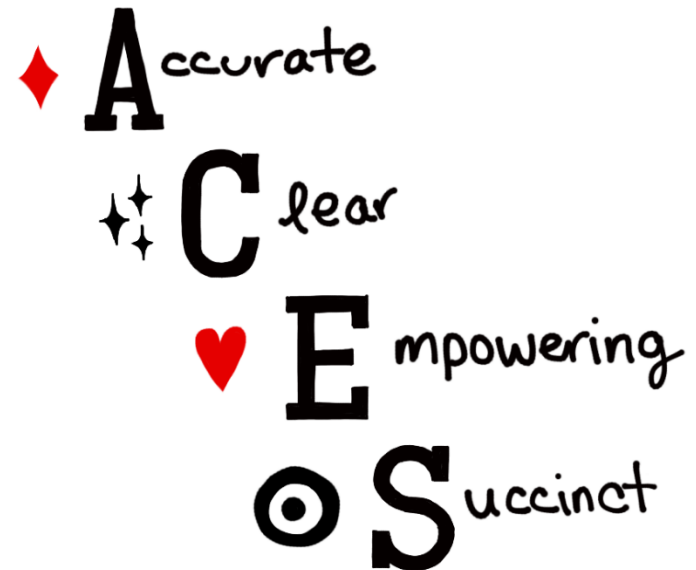
Descriptive title, axes, labels

Layout

Remember about patterns

Empowering

Succinct



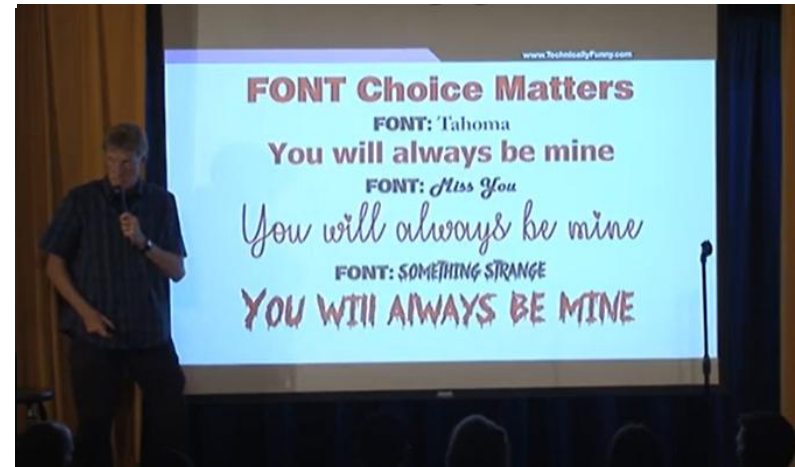
Font matters

Many of us take font choice for granted.

The font you use is part of your message

Fancy Chart Title ❌

Legible Chart Title ✅



<https://www.youtube.com/watch?v=UZBC7maSOX4>

Colour

Differentiate groups with distinct shades
(color wheel)

Avoid colors that are too bright
(reduce eye strain)

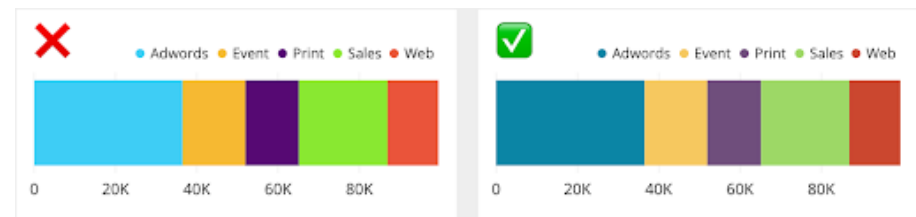
Keep all colors within a similar range of
brightness and saturation

*(so that one group does not stand out too much
from the others)*

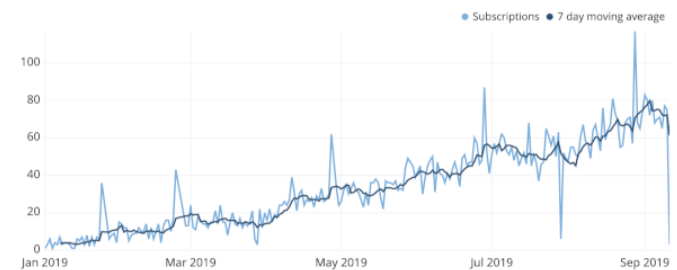
*The exception is related groups of data, for
example, daily data values next to a moving
weekly average.*

If you move the same data or groups between
charts, the colors used should be consistent.

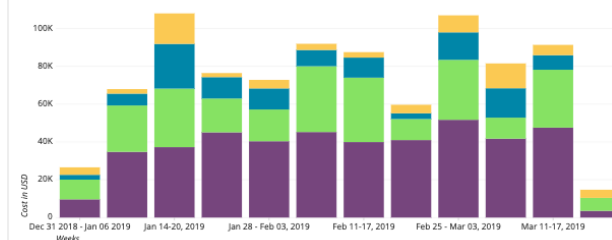
This makes it easy to create connections in your
data between charts, tables, and charts.



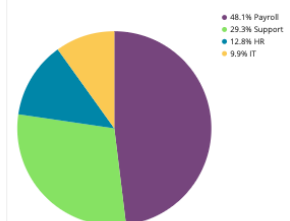
New Subscriptions



Operations Costs over Time by Department



Operations Cost by Department



Context

Always include information such as:

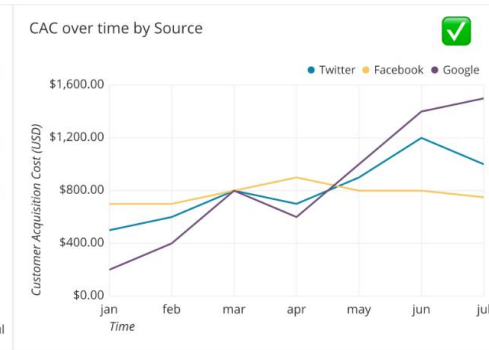
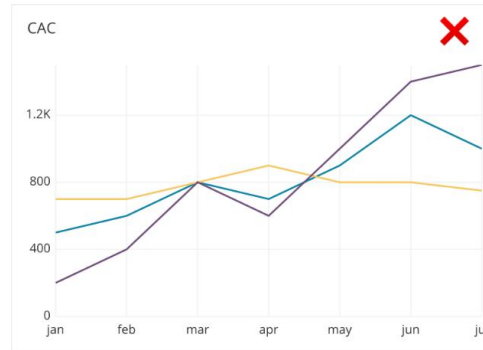
Descriptive title

Descriptive axis titles

Categorical labels

If the data is difficult to decipher, it will not be read.

The more explanation or context needed to understand the charts on the dashboard, the less effective the dashboard will be as an analytics tool.



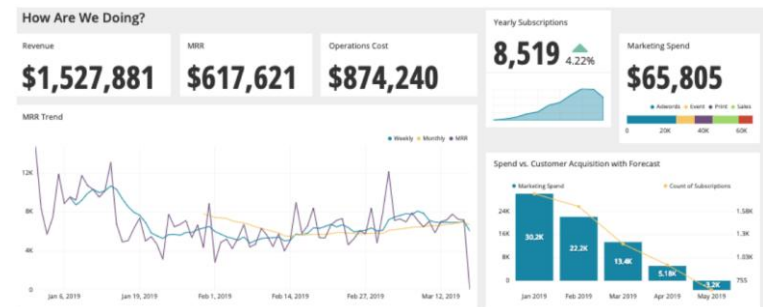
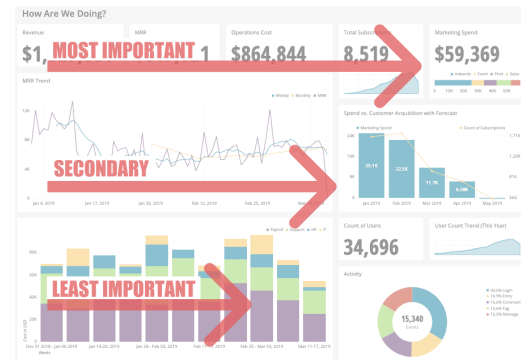
Arrangement

Users in most countries read from left to right and top to bottom.

Therefore, the most important information should be in the upper left corner and the least important information in the lower right corner.

Visuals should be aligned.

A chart that stands out from other visuals will distract from the goal of clearly presenting all the information to the viewer.



An optimal dashboard is ACES:

Accurate

Clear

Empowering

*Does a dashboard get used regularly
and does it help people make
decisions?*

Do they view it regularly?

Keep dashboard performance
metrics

Does it factor into their decision
making?

A simple way to check this is to ask
the viewer: if the numbers on this
dashboard went to 0 or if the
numbers doubled, would you do
anything?

Succinct

♦ **A**ccurate
✦ ✦ **C**lear
♥ **E**mpowering
◎ **S**uccinct

An optimal dashboard is ACES:

Accurate

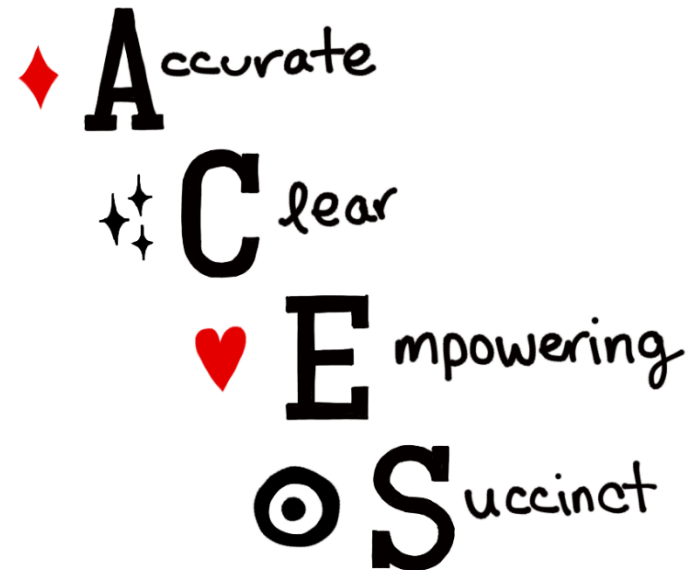
Clear

Empowering

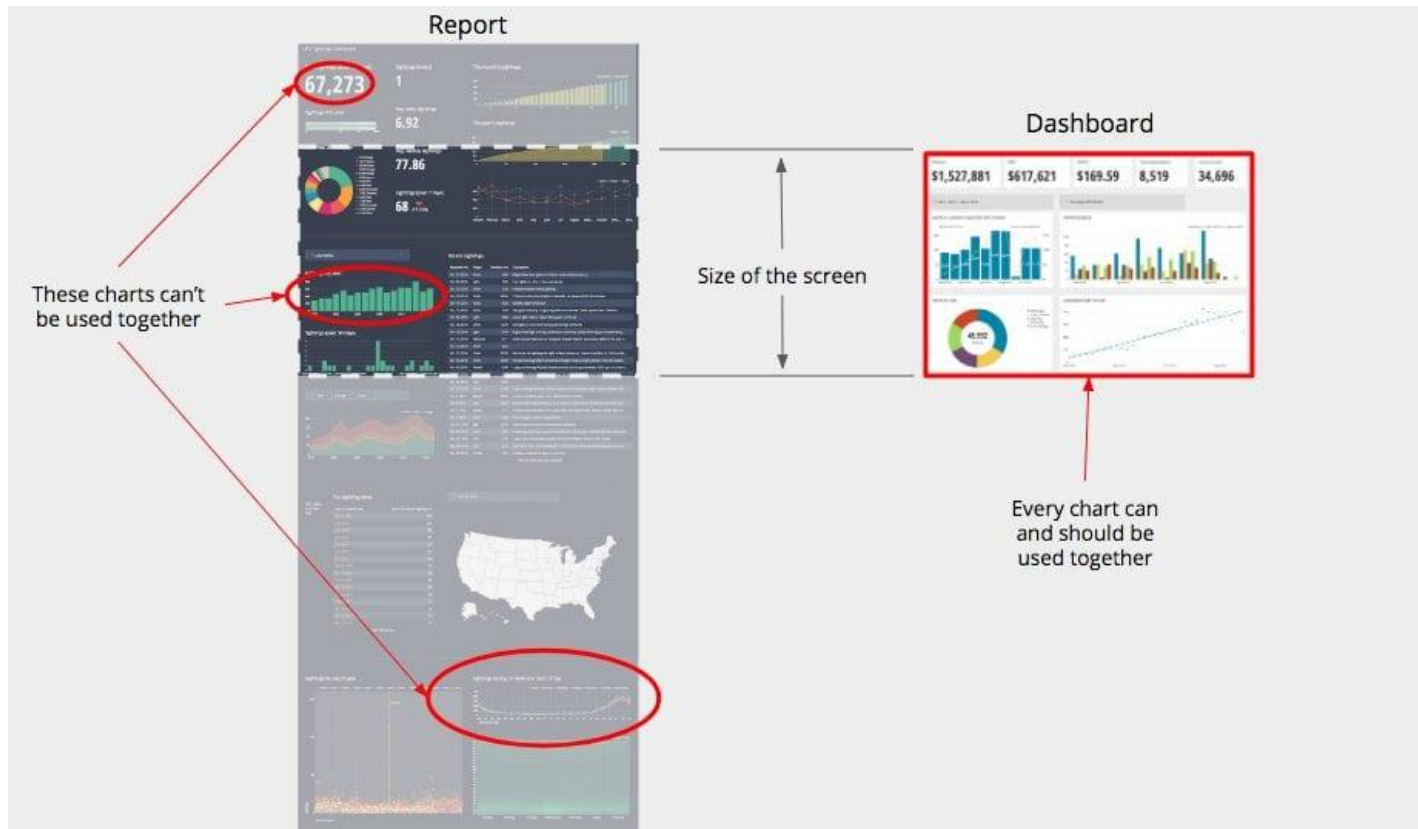
Succinct

One of the main benefits of a dashboard is that it shows multiple data visualizations simultaneously.

This helps facilitate processing of all of the information together.

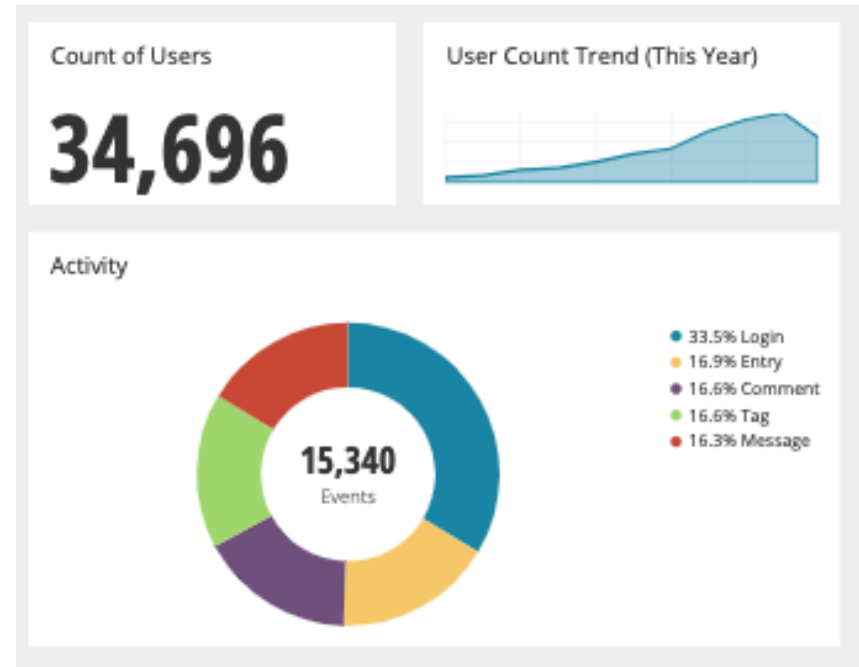


Scrolling becomes counterproductive



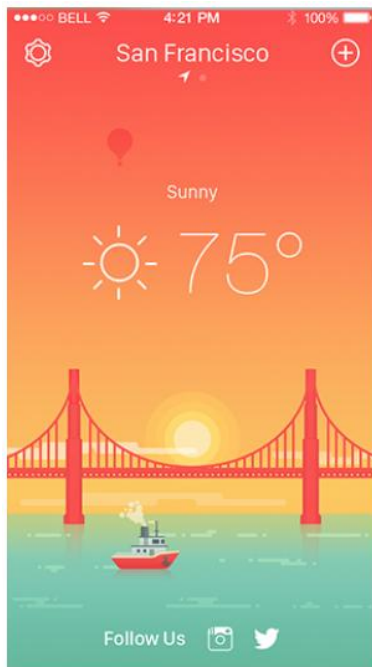
Proximity

Data that is relevant to other pieces of information on the dashboard should be placed in close proximity on your dashboard.

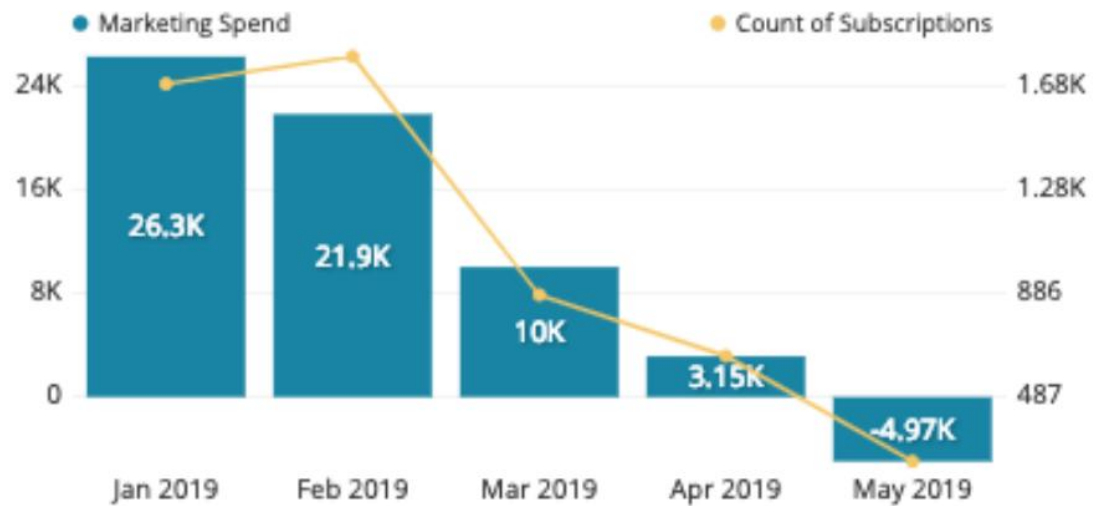


Proximity

Random data shouldn't go together



Spend vs. Customer Acquisition with Forecast



The importance of KPIs

KPIs are an essential part of dashboard architecture.

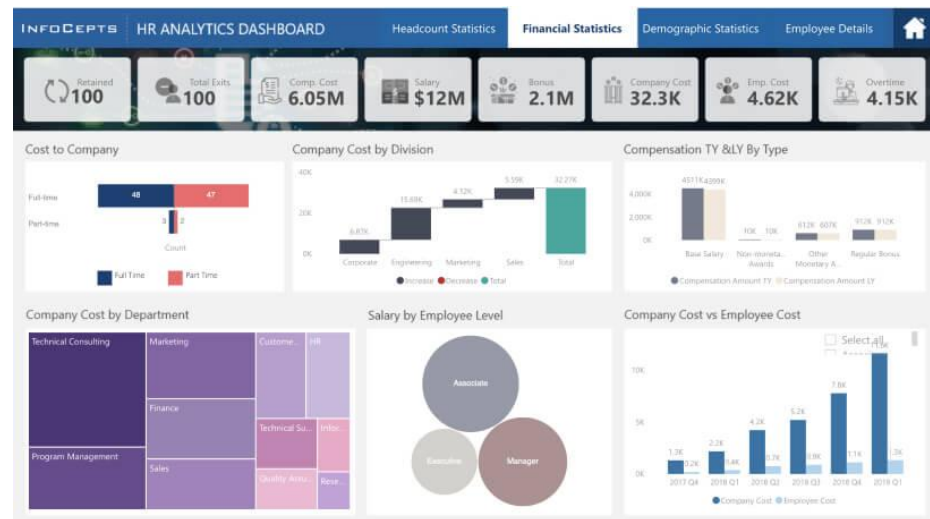
You need to know where you are now in order to know the best step to take next.

When choosing your dashboard's KPIs

select only the ones that help answer your primary question

not so many that it dilutes your data,

but not so few that you don't get an accurate picture.



Think ACES

Accurate

If the data that is visualized is incorrect or the visualization biases how it is interpreted, the dashboard becomes unusable

Clear

Clarity allows for increased speed of insights

Empowering

Will people access the dashboard regularly to make decisions
the dashboard delivers on its goal of supporting decisions

Succinct

Keeping a dashboard brief and relevant to critical information
makes it easy to evaluate all of the data simultaneously and make decisions

1



Design
for a target

2



Keep everything
at a glance

3



Keep it simple

4



Align elements

5



Be consistent

6



Highlight the most
relevant information

7



Be clear

8



Start from zero

9



Shorten the
numbers

10



Show the
context

11



Choose the
right colors

12



Design dashboards,
not reports

13



Show variations

14



Leave the
noise off

15



Pick the
right charts

Dashboard mistakes to avoid

Avoid these common data dashboard pitfalls:

Not creating a clear goal for the dashboard

Implementing too many KPIs per dashboard

Using low-quality data

Representing data in the wrong type of graph or chart

Presenting too much information in one dashboard

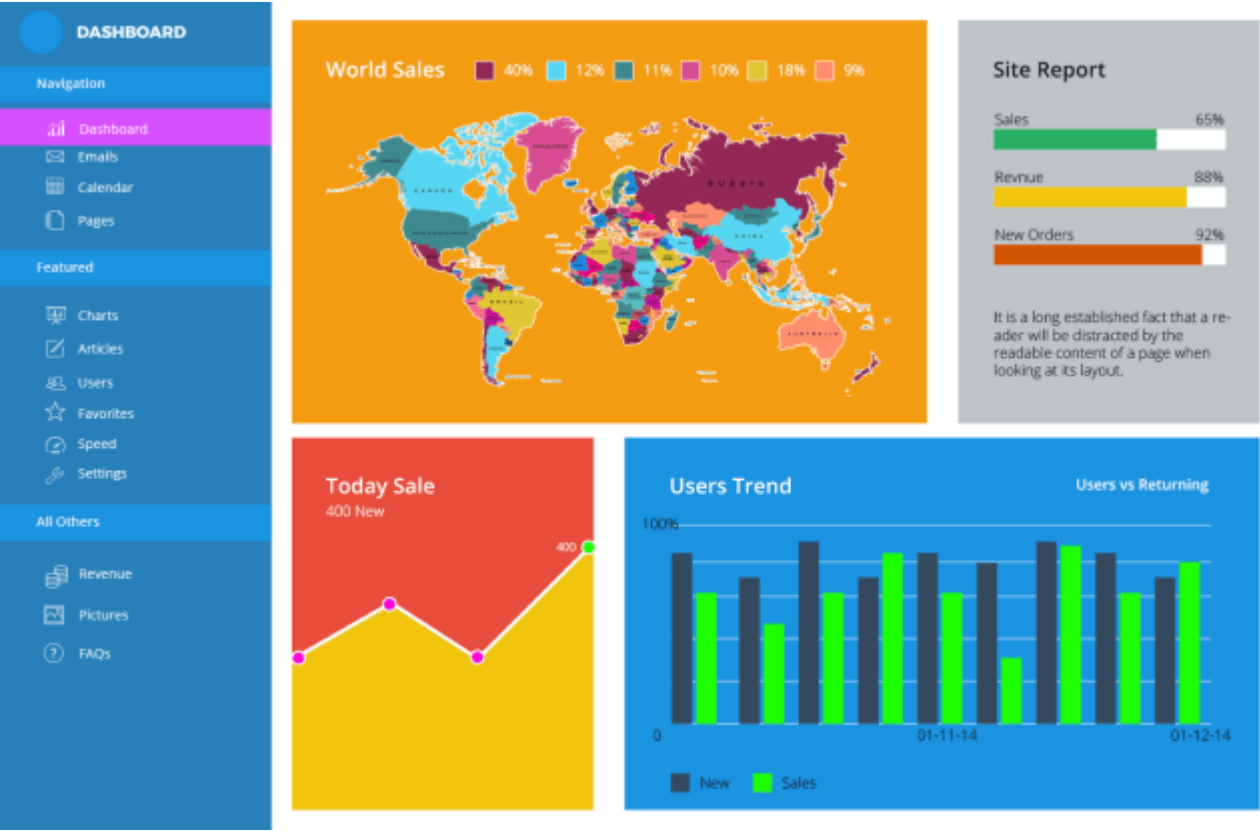
Using a cluttered, confusing, or inaccessible design



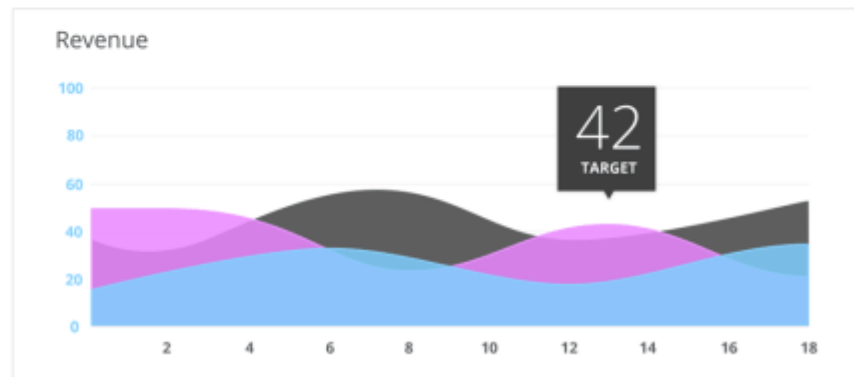
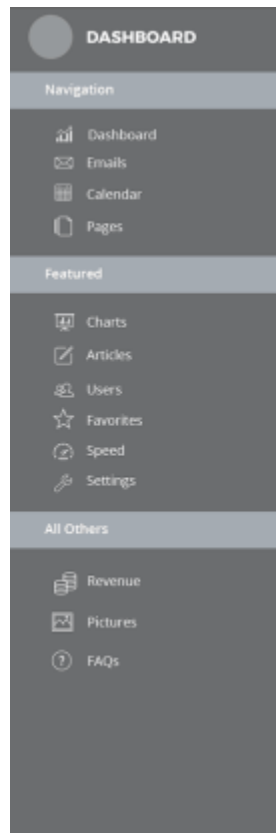
Too Much Clutter and Unnecessary Variety



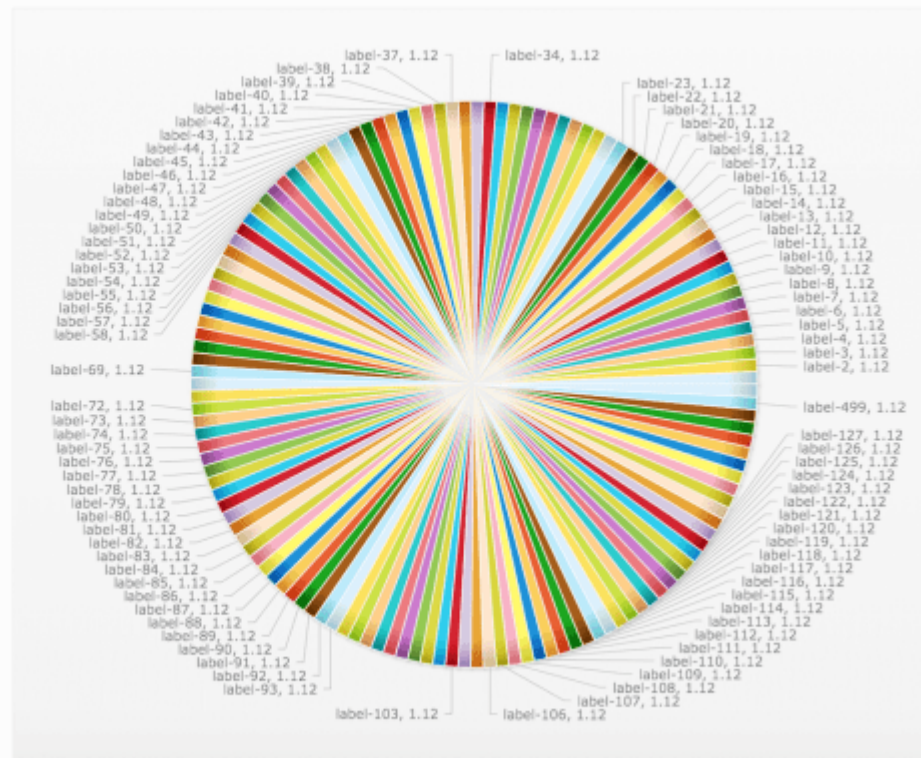
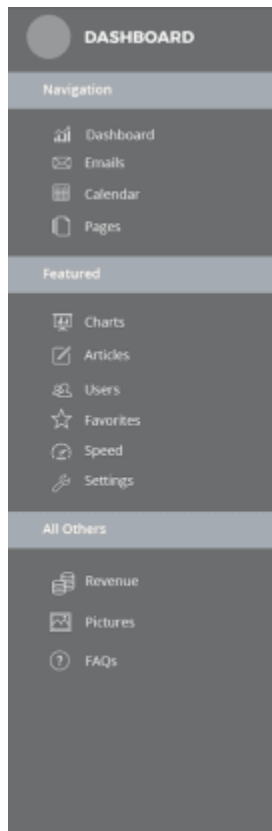
Too Many Colors



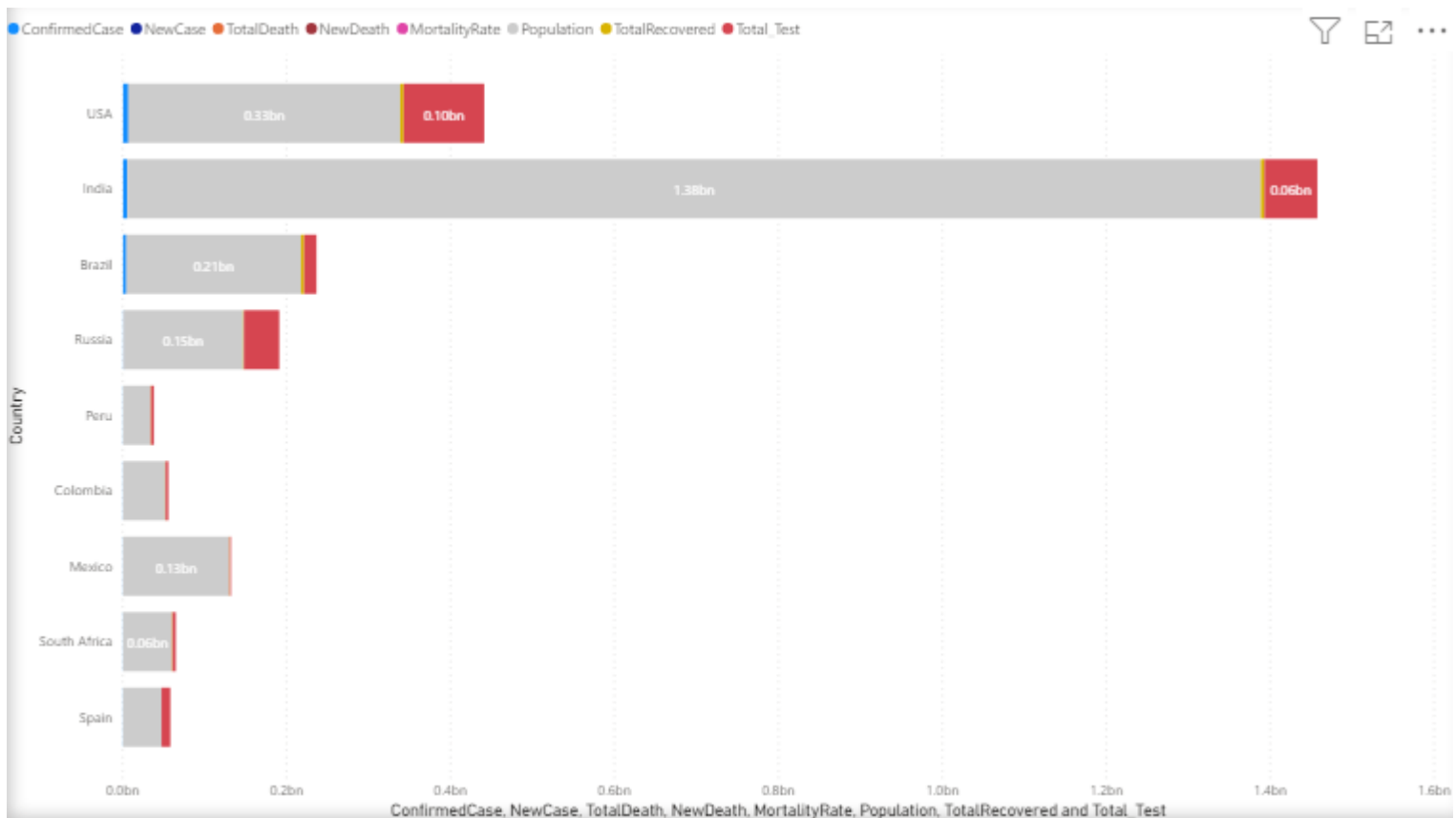
Lack of Context



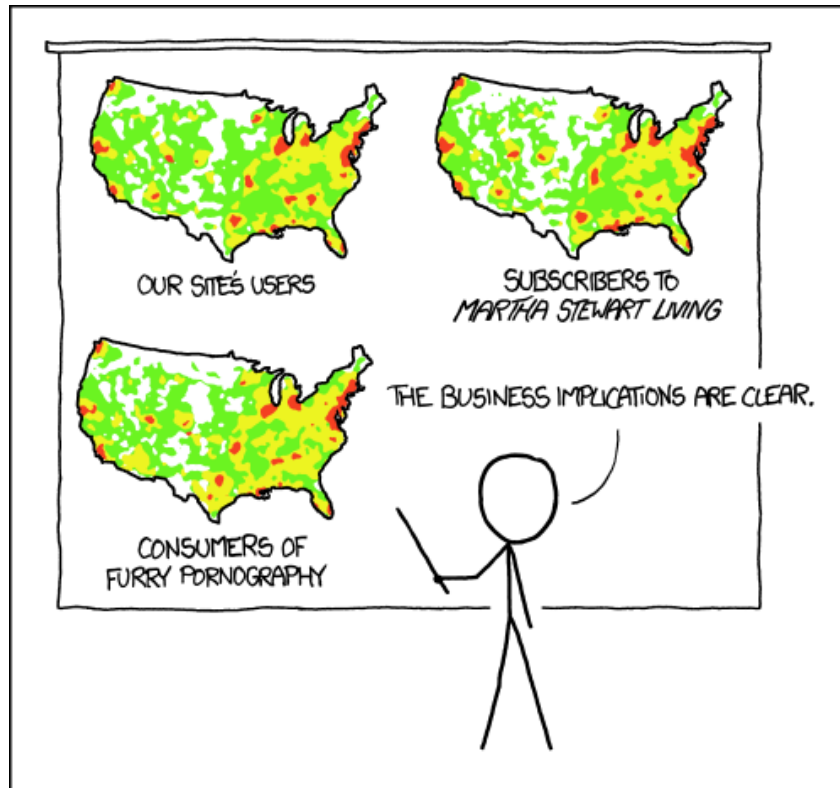
Bad Data-to-Visualization Pairing



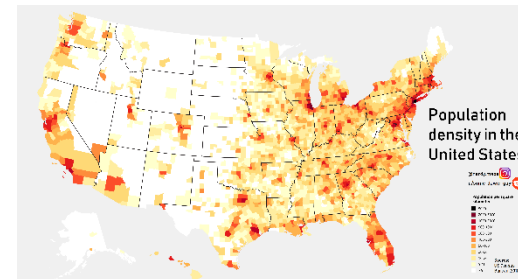
Information congestion is another example of bad data visualization



In the end – interpretation



PET PEEVE #208:
GEOGRAPHIC PROFILE MAPS WHICH ARE
BASICALLY JUST POPULATION MAPS



G. Zelazny

The say it with charts – complete toolkit
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